The Making of Dominant Vehicle Currencies

August 19, 2026

Abstract

Transaction-level decentralised-exchange routes reveal which asset carries each indirect trade. Stablecoins' route share rises from 16.9% to 42.3% between the first halves of 2024 and 2026, but matched pairs show little stable-for-native substitution; reallocation across markets carries the shift. Mechanism evidence points to market formation rather than only within-pair switching: entrants adopt stable vehicles, stable-born pairs stay stable, USDC and USDT dominate stable entry, local two-leg bridge depth predicts route choice, and stable route use forecasts bridge deepening. Liquidity provision moves more slowly: stable-route demand rises during USDC/SVB while stable capital falls.

1 Introduction

The U.S. dollar appears on more than 85% of foreign-exchange transactions, far more often than the United States' share of cross-border economic activity would suggest (Somogyi, 2026). Its use as a vehicle between other currencies helps explain this prominence. Vehicle use concentrates trading in dollar markets; deeper markets lower trading costs; and lower costs reinforce the dollar's intermediary role (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). This feedback explains why an established vehicle can persist. It says less about how dominance changes when new markets and trading technologies alter the routes available to traders.

The distinction is practical. A treasury desk exchanging two thin currencies may route through dollars because the direct market is expensive. If that same desk keeps using dollars, the dollar's role inside that corridor has not changed. But the aggregate dollar share can still fall if trade grows fastest in another corridor that already settles or routes on different terms. A vehicle-currency statistic is therefore a statement about two objects at once: the intermediary chosen inside each corridor and the distribution of trading across corridors. The first is the margin classical vehicle-currency theory emphasises. The second is visible only when the data preserve enough of the route to say which ultimate exchange each intermediary served.

Testing how dominance changes requires observing the full trading path. Standard foreign-exchange records report turnover by currency pair, not whether two reported trades belong to the same underlying exchange.[1] A yen–euro exchange routed through dollars therefore appears as

[1] For example, see the BIS 2025 Triennial Survey reporting guidelines and templates.

turnover in the yen–dollar and dollar–euro markets, with no link between the two legs. Somogyi (2026) overcomes this limitation by using pair-level volume responses around non-overlapping holidays to infer vehicle trading. We take a complementary approach in a market where transaction logs preserve the sequence of pools and assets used in each exchange.

Decentralised exchanges split liquidity across smart-contract pools that a router can combine within one transaction. The path is observable. We reconstruct those combinations from 472 million swaps on eight Ethereum automated market maker (AMM) deployments between February 2020 and June 2026. The reconstruction identifies the first and last assets in the observed pool sequence, every intermediary, and any movement across exchanges. It thus separates intermediary use from use at either end of the observed route. These pool-route endpoints need not coincide with the user's original instruction, a distinction we document with a transaction in Section 2. This route-based measure complements measures of quoting, invoicing, and funding, which describe related but different currency choices (Flandreau and Jobst, 2009; Gopinath et al., 2020; Gopinath and Stein, 2021).

Stablecoins become substantially more prominent vehicles during the sample. Comparing the same January-through-June dates in 2024 and 2026, their share among native and stable intermediaries rises from 16.9% to 42.3% by route count and from 32.7% to 76.5% by dollar-weighted volume. The set of traded markets carries the aggregate shift; matched markets contribute little. Holding the ordered source–destination pair, the calendar date, and the observed route scope fixed, stable-for-native substitution is +0.2 pp by count, with a standard error of 0.8 pp, and −1.3 pp by value, with a standard error of 2.2 pp; both intervals exclude changes of anything approaching the aggregate size. The aggregate accumulates instead from two composition margins: routed activity reallocates toward continuing pairs that already use a stablecoin intermediary, and pairs observed in only one of the two years supply the remainder. The transition is also not monotone. Native assets regain share in 2023 and 2024, stablecoins recover the value lead in 2025, and their lead by route count is temporary. The change is especially pronounced for USDT, whose intermediary use rises markedly relative to its appearance at observed route endpoints. Stablecoin share rises separately among single-exchange and cross-exchange routes, even as cross-exchange routing expands rapidly.

Classical vehicle-currency models ask which intermediary minimises the cost of exchanging a given pair of assets. Their volume–cost feedback explains why trading converges on a common vehicle and why an established vehicle persists (Krugman, 1980). The route decomposition adds a second empirical margin: the distribution of activity across source–destination pairs. The distinction is not peculiar to this market. China and Brazil have established clearing arrangements for settling their bilateral trade in renminbi, and a dollar share computed across all trade corridors declines as such corridors grow even when every corridor that settles in dollars today continues to do so.[2] Relative execution costs and routing software can change the intermediary selected for a

[2]See the memorandum of understanding between the People's Bank of China and the Banco Central do Brasil on establishing renminbi clearing arrangements in Brazil. The comparison is illustrative. We measure no currency's share of trade settlement and make no claim about the arrangement or its consequences.

given pair, while liquidity supply, asset risk, and protocol design can alter the routes through which activity occurs. A satisfactory explanation for the stablecoin gain must account for the reallocation across pairs that carries most of it.

We make three contributions. First, transaction sequences directly measure a currency role usually inferred from aggregate turnover. Linking the legs of an exchange separates intermediary use from use at the observed route endpoints without an exclusion restriction for latent vehicle trades, complementing the holiday-based inference of Somogyi (2026) and the historical turnover evidence of Flandreau and Jobst (2009). Second, we show that aggregate vehicle dominance depends on the allocation of routed activity across source–destination pairs as well as intermediary choice within matched pairs. Third, we document how the new vehicle role is made: stable vehicles appear at market birth and as native-only corridors accumulate repeated route demand, USDC and USDT account for nearly all stable-entry routes, entry candidate identity persists, local two-leg bridge depth predicts route share beyond candidate reach including at market birth, stable route use forecasts later bridge deepening, and provider capital adjusts asymmetrically to route-capital imbalances and peg stress. These results connect the self-reinforcing depth mechanism in vehicle-currency theory to research on fragmented trading and automated market making, where market design and liquidity supply shape the opportunities available to traders (Chen and Duffie, 2021; Lehar and Parlour, 2025).

The remainder of the paper proceeds as follows. Section 2 describes the exchanges, reconstructs transaction-level routes, and defines vehicle dominance. Section 3 documents the rotation toward stablecoins and separates changes across pairs from substitution within matched pairs. Section 4 defines the counterfactual cost benchmark and the liquidity it represents. Section 5 tests the route evidence against path complexity, exchange span, the software that assembles a path, and the pricing rule applied at each leg, and relates it to liquidity supply and protocol architecture.

2 Setting and data

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine atomically. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Every call either settles in the same transaction or reverts with the rest of the route.[3] This design makes the intermediary sequence observable. A route through an intermediary is unavoidable when no direct pool connects the assets; when direct and indirect pools coexist, the router compares their fees, price impact, gas, and venue access. Lehar and Parlour (2025) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a

[3] Ethereum treats a transaction as one state transition and records its execution status and ordered logs in the transaction receipt; see the Ethereum Yellow Paper and the official Ethereum JSON-RPC documentation.

split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction. Throughout, for a reconstructed sequence $A \to B \to C$, we call $A \to C$ the *ultimate trade* and (A, C) its ordered *ultimate pair*. We call $A \to B$ and $B \to C$ the *atomic trades*, and their ordered token pairs the *atomic pairs*; *leg* is shorthand for one atomic trade. Ultimate endpoints belong to the connected pool component and need not coincide with the endpoints of a broader user instruction.

A transaction on January 10, 2026 illustrates both the information in the logs and its boundary. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.[4] The user's instruction begins with PYUSD, but the exchange-pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The route observed in the pools is therefore USDC–USDT–USDe: its ultimate trade is USDC→USDe, its atomic trades are USDC→USDT and USDT→USDe, and USDT is the vehicle. We do not assign the earlier PYUSD–USDC conversion to an exchange pool because it occurs outside this connected sequence. Applying the same rule throughout the sample preserves the distinction between the user's instruction and the trading path we can observe.

The sequence directly identifies intermediary use. Conditional on reconstructed pool states immediately before execution, it also defines a same-size cost comparison between the observed path and available alternatives. We introduce notation for the route here and develop the cost comparison in Section 4.

Write $\mathcal{A}_t$ for the assets tradable on day t and $\mathcal{P}_t$ for the pools live on that day. For a route r, let $e(r)$ record its block, transaction position, and first pool-event position. The notation $e(r)^-$ refers to the state immediately before that transaction begins; a daily closing state generally occurs later. A route carries an ordered pool sequence $(p_1, \ldots, p_{m(r)})$ and token sequence $(k_0, \ldots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. Thus (i, o) is the ultimate pair and each (k_{h-1}, k_h) is an atomic pair. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p's state immediately before execution and let $Q_p(i \to j, \Delta_i; \chi_{p,e^-})$ denote the units of token j returned for an exact input of Δ_i units of token i. Composing these pool quotes gives the route's token output and dollar value,

$$\begin{aligned} \Delta_{k_h} &= Q_{p_h}\left(k_{h-1} \to k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}\right), & h &= 1, \ldots, m(r), \\ Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o,e(r)} Q_r(q, e(r)). \end{aligned} \tag{1}$$

The venue-specific forms of Q_p appear in Appendix B. Pool invariants take token quantities as arguments, which is why the dollar notional q and the token input Δ_i remain distinct. Equation (1) describes an executed route and, once market state has been reconstructed, a hypothetical route.

Intermediary use is one of several currency roles. Invoicing concerns the currency named in trade contracts (Gopinath et al., 2020; Amiti et al., 2022; Mukhin, 2022); funding concerns the denomination of liabilities (Gopinath and Stein, 2021; Eren and Malamud, 2022); and dealer turnover records trading activity (Somogyi, 2026). These roles can reinforce one another. The path in

[4] Transaction `0xbda1d07f…9bf67ad9`.

Equation (1) identifies the asset used between the currency sold and the currency bought.

This route setting makes the classical volume–cost mechanism observable at transaction level. Greater use of a vehicle expands trading in the two markets joined through it; deeper markets can then lower the cost of using the same vehicle again (Krugman, 1980). In Kiyotaki and Wright (1989), traders accept an asset for indirect exchange when its marketability compensates for its storage cost. Dowd and Greenaway (1993) emphasise the fixed costs of learning, accounting, and quoting in a new currency. Liquidity placement, venue access, and the routes available to software provide corresponding coordination margins on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route consists of two atomic trades, one on either side of a unique intermediary. This gives the primary measure a simple unit: each ultimate trade contributes one intermediary observation. A supplementary multiple-intermediary measure includes routes of any length and counts each intermediary use separately. For route r, let $\mathcal{M}(r)$ denote the set of assets between the ultimate source and destination. We record the intermediary status of asset k as

$$Z_{r,k} \;=\; \mathbf{1}\{k \in \mathcal{M}(r)\}. \tag{2}$$

Let $a(k)$ assign each token to an economic asset type. The native group consolidates ETH and its one-for-one wrapped representation, WETH, after route reconstruction. The stable group contains fiat-reserve, on-chain-collateralised, synthetic, and non-dollar stable currencies. For $g \in \{\text{native}, \text{stable}\}$, define $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$; on an exact two-leg route, this sum is an indicator.

We begin with the competition between native and stable intermediaries. Let $\mathcal{R}_t^{(2),NS}$ denote exact two-leg routes on day t whose intermediary belongs to either group, and let $\mathcal{R}_t^{(2),NS,V}$ retain routes for which the source, intermediary, and destination values agree within 20%. The equally weighted and dollar-weighted group shares are

$$S_{g,t}^{N} \;=\; \frac{\sum_{r\in\mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \qquad S_{g,t}^{V} \;=\; \frac{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r\in\mathcal{R}_t^{(2),NS,V}} \nu_r}, \tag{3}$$

where ν_r is the arithmetic mean of the route's total dollar value at its observed source and destination. The 20% rule also requires every intermediary's incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. Thus $S_{\text{stable},t}^{N}$ is stablecoins' aggregate share among native and stable intermediaries, and $S_{\text{stable},t}^{V}$ is the corresponding dollar-weighted share. The two groups exhaust this denominator.

The frequency with which an asset appears at the beginning or end of an observed pool route provides a benchmark for its intermediary use. This comparison uses a broader set of currencies $\mathcal{C}$: native, staked-native, stable, and imported currencies, with unclassified assets excluded. We retain every complete route that does not return to its starting currency. Direct routes contribute to the source and destination counts, and a longer route contributes one observation for each intermediary it uses. Let $n^I_{k,t}$ count intermediary uses and $n^E_{k,t}$ count source or destination appearances for $k \in \mathcal{C}$. The two shares are

$$I^N_{k,t} = \frac{n^I_{k,t}}{\sum_{j\in\mathcal{C}} n^I_{j,t}}, \qquad E^N_{k,t} = \frac{n^E_{k,t}}{\sum_{j\in\mathcal{C}} n^E_{j,t}}, \tag{4}$$

and the corresponding share gap and excess-use ratio are

$$\Delta^N_{k,t} = I^N_{k,t} - E^N_{k,t}, \qquad X^N_{k,t} = \frac{I^N_{k,t}}{E^N_{k,t}} \quad \text{when } E^N_{k,t} > 0. \tag{5}$$

A positive $\Delta^N_{k,t}$ means that the asset's share of intermediary use exceeds its share of source and destination appearances; $X^N_{k,t}$ expresses the same comparison as a ratio. The dollar-weighted measures replace counts with route values while retaining the same set of currencies and the 20% agreement rule. Because excess use compares each currency's intermediary and source-or-destination roles, its denominator differs from the native-or-stable denominator in Equation (3). The latter remains the measure of the headline rotation.

Source and destination assets must differ in the analysis sample. We classify routes that return to their starting asset as round trips and remove them. Such paths may contain cyclic arbitrage or other self-returning activity: geometry identifies the cycle, although trader intent remains unobserved. Daian et al. (2020) show that several arbitrage trades can be bundled atomically. On the median of 79 sampled days, round trips account for 12.7% of multi-leg routes by count and 21.7% by value. Appendix A reports their distribution across the sampled dates.

Within this route sample, we classify assets by the claims they represent: native, staked native, stable, imported, and other. The native category contains the platform's settlement asset. We treat ETH and its one-for-one wrapped representation, WETH, as one economic asset after reconstructing the route. This consolidation is an economic classification, not an address-level identity (Milionis et al., 2022). Liquid-staking claims remain separate because they also convey staking returns. Stable assets target a fiat value, imported assets are wrapped claims on off-chain or cross-chain value, and the remaining assets enter the residual category; the economic claim represented by an asset determines its category. Stable assets themselves encompass economically different designs. Reserve-backed stablecoins rely on backing and redemption, making confidence in those arrangements relevant to acceptance and run risk (Gorton and Zhang, 2023); fiat-backed, crypto-collateralized, and algorithmic stablecoins also differ in their solvency and peg-restoration mechanisms (Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). Section 3.3 therefore examines the two largest reserve-backed stablecoins separately.

2.3 The route panel

Our route panel covers Curve, Uniswap v2, v3 and v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. Table 1 summarises its coverage. The sample spans 2,332 calendar dates from February 11, 2020 through June 30, 2026. These deployments include several major exchanges but remain a subset of Ethereum trading. Protocols enter after their launch dates, so the exchanges and routes available to traders expand over the sample. We keep 55 early dates with no observed swaps as zero-activity dates for the covered deployments.

We begin with 472,254,909 pool-level swap records. The Graph supplies indexed events for seven exchange series, Dune's `dex.trades` table supplies Fluid records, and Ethereum JSON-RPC supplies block headers, ordered logs and receipts, token precision, factory registries, and state checks.[5] After harmonising trade direction, event identity, and token units, the panel contains 471,616,269 directed atomic trades (legs). We order those atomic trades within each transaction and apply the flow rule above to distinguish single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction algorithm and reports the historical token units that we cannot verify.

Every reconstructed token sequence enters the count measure. Dollar-weighted results use routes for which the source, intermediary, and destination values agree within 20%. Keeping the count sample broader preserves observed intermediary choice when token valuation is noisy.

Transactions and contract-state changes are publicly observable (Schär, 2021). Reconstructing a route still requires matching pool events, venues, and token units. This procedure identifies the executed route and its calling contract, but an executor address does not establish which frontend or software formed the quote. Existing router labels likewise cover only part of decentralised-exchange activity (Xi and Moallemi, 2026). On a 2024 snapshot of 74,323 swaps, 241 distinct senders appear, and a registry of the ten largest accounts for 67.7% of those swaps. By a 2025 snapshot, 397 senders appear across 130,282 swaps and the same registry covers 11.8%. We therefore report route-level results without attributing them to quote authors.

Table 1: The route panel

	Route panel
Calendar partitions	2,332
Missing source-days	0
Raw swap rows	472,254,909
Usable directed atomic trades (legs)	471,616,269
Routes per day, median of 79 sampled days	161,152
Multi-leg routes per day, median of 79 sampled days	31,253
Multi-leg share of routes, median of 79 sampled days	19.7%
Round trips, share of multi-leg routes by count	12.7%
Round trips, share of multi-leg routes by value	21.7%

Note: Calendar partitions are UTC dates in the full eight-deployment panel from 11 February 2020 through 30 June 2026. Raw swaps and usable legs use the full panel. Daily route statistics and round-trip shares use 79 sampled dates from 1 June 2020 through 27 April 2026. A multi-leg route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-leg routes.

[5]See The Graph's subgraph documentation, Dune's curated-data documentation, and the official Ethereum JSON-RPC documentation.

Table 2: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change (s.e.)
Route count	16.9%	42.3%	+25.4 pp (1.05 pp)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.9 pp (2.02 pp)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable; the reported share aggregates all stablecoins in that denominator. Daily shares receive equal weight over the days of the calendar year observed in both years. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

3 Rotation in the vehicle role

3.1 A route-level transition

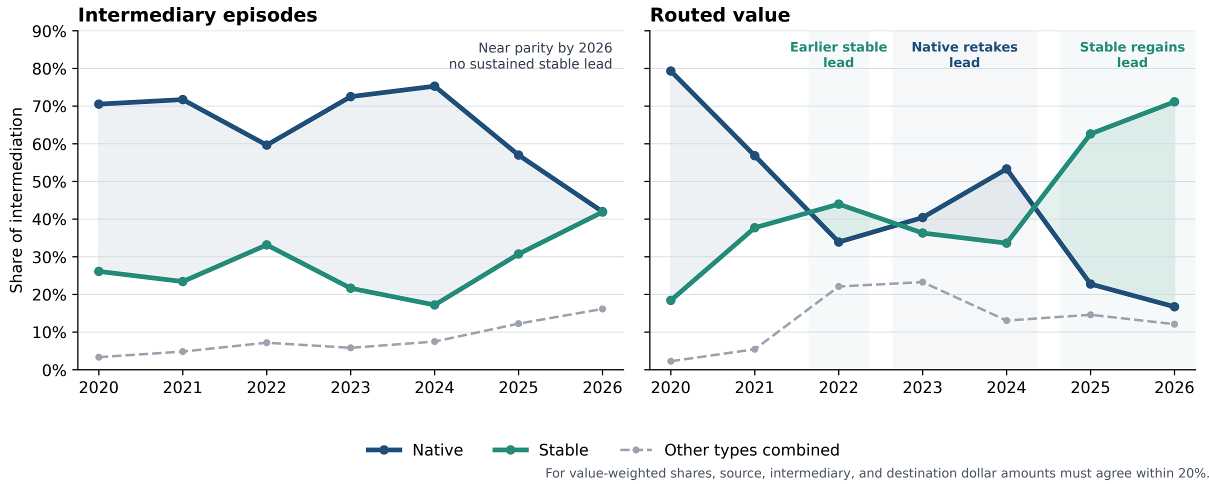
Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. Comparing the same days of the calendar year, their share among native and stable intermediaries rises from 16.9% to 42.3% by route count. The paired-day change is +25.4 pp, with a standard error of 1.05 pp. Their dollar-weighted share rises from 32.7% to 76.5%, an increase of +43.9 pp with a standard error of 2.02 pp. The count series gives equal weight to each exact two-leg route. The dollar-weighted series uses routes for which the values at the source, intermediary, and destination agree within 20%. Table 2 reports both. Each figure is an average over the markets traded in its year, and those markets differ between the two years. Section 3.2 shows that the difference is what moves the aggregate: within a source–destination pair traded in both years, the intermediary carrying the route barely changes.

Figures 1 and 3 extend the comparison to routes of any length and all intermediary types. Each intermediary used on a longer route contributes one observation. In 2026, the 20% value-agreement sample covers 71.6% of native intermediary value and 55.1% of stablecoin intermediary value. The annual path is not monotone. Native assets regain share in 2023 and 2024, stablecoins recover the dollar-weighted lead in 2025, and their lead by route count is shorter lived.

The denominator so far treats a stablecoin as a stablecoin. What backs one moves over the sample, because the collateral an issuer may hold is itself revisable, so each regime label carries a date. Of the 21 stablecoins observed as intermediaries here, 2 change regime within the sample and 0 change between the two years compared. Cutting the same share on the same denominator, calendar, and routing strata by the intermediary's dated regime therefore holds each regime's membership fixed across the comparison, and the cut divides the class the way the stablecoin literature does, by what stands behind the peg.

Fiat-reserve claims move +24.50 pp of the +25.42 pp count rotation, with a standard error of 1.08 pp, and +44.09 pp of the +43.87 pp dollar-weighted rotation, with a standard error of 2.07 pp. The regimes divide one denominator, so their changes add to the class's, and the residue the fiat-reserve term leaves is small and mixed in sign. Synthetic claims multiply their share of the class's routes from 0.3% to 1.4% while contributing +0.55 pp of the count rotation and +1.66 pp

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete routes that do not return to their starting asset. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Table 2 uses the narrower native-or-stable denominator. The 2026 observation covers January through June.

of the dollar-weighted one. Claims that mix real-world assets into on-chain collateral fall −1.73 pp by value, from 6.8% to 1.4% of the class's dollars. The rotation is therefore a movement into dollar tokens backed by reserves; the class's fastest-growing regime moves the aggregate by under two points on either measure. The cut is measured on the released endpoint-candidate panel; against the daily type panel the pooled rotation differs by 0.001 pp by count and 0.000 pp by value, so the regime terms sit beside the headline in the same units.

3.2 Composition of ultimate pairs

A vehicle's aggregate share can rise in three ways. It can be chosen more often within an ultimate (source–destination) pair traded in both years; routed activity can move toward continuing ultimate pairs that already use it; or ultimate pairs traded in only one of the two years can route through it. Table 4 shows that the first way accounts for little of the stablecoin increase and that the other two carry it.

Table 2 averages daily shares, whereas Panels A through C of Table 4 pool route count or value across the matched January–June windows. The different weighting explains why the headline changes of +25.4 pp and +43.9 pp differ slightly from the decomposition totals of +25.7 pp and +42.8 pp.

Those three ways form an accounting identity once the third is split in two, because pairs traded in only one year both route their activity differently and carry a changing share of it. The rest of this section reads the resulting four terms one at a time. Fix one of the measures in Equation (3) and let p index ordered source–destination pairs. A pair is continuing when it carries native-or-stable routes in both comparison years and year-specific when it carries them in only one. Let W_y be the share of year-y native-plus-stable mass carried by continuing pairs, $\omega_{p,y}$ the weight of continuing pair p within that mass, and $s_{p,y}$ its stablecoin share, so continuing pairs route $S_{\text{com},y} = \sum_p \omega_{p,y} s_{p,y}$ of their activity through a stablecoin and year-specific pairs route $S_{\text{exc},y}$ of theirs. The aggregate share is $S_y = W_y S_{\text{com},y} + (1 - W_y) S_{\text{exc},y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$,

$$\Delta S = \underbrace{\bar{W} \sum_p \bar{\omega}_p \, \Delta s_p}_{\text{within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \, \Delta \omega_p}_{\text{reweighting}} + \underbrace{(\bar{S}_{\text{com}} - \bar{S}_{\text{exc}}) \Delta W}_{\text{support mass}} + \underbrace{(1 - \bar{W}) \, \Delta S_{\text{exc}}}_{\text{year-specific pairs}} . \tag{6}$$

Midpoint weights make the identity exact by splitting each product's interaction evenly between its two factors instead of charging it to one year. Panels B and C report the four terms. Within continuing pairs, the stablecoin share moves −0.1 pp by route count and 0.0 pp by value. Reweighting across those pairs contributes +8.6 pp and +26.2 pp; the migration of mass between the continuing and year-specific populations, −0.5 pp and −2.5 pp; and the pairs traded in only one year, +17.8 pp and +19.2 pp. The first of the three ways is worth almost nothing on either measure, and the other two carry the rotation between them. Somogyi (2026) decomposes dollar dominance in the other direction, separating the volume of a given dollar pair into the part that exists to route two other currencies and the part that does not. That split runs inside a pair; the one that carries the rotation measured here runs across them.

Panel A factors the same route-count total a second way. Equation (6) takes a pair's vehicle-routed activity as given and asks only which vehicle carried it; Panel A asks first how much the pair traded, then how much of that trading passed through any intermediary at all, and only then which intermediary it was. Let $M_{p,y}$ be the routes observed in pair p in year y, let $I_{p,y}$ be the fraction of them that passed through a native or stablecoin intermediary, and let $s_{p,y}$ be the stablecoin share of that intermediated activity, the same object the identity above assigns to a continuing pair. For any set of pairs $\mathcal{Q}$,

$$F_y(\mathcal{Q}) = \frac{\sum_{p \in \mathcal{Q}} M_{p,y} I_{p,y} s_{p,y}}{\sum_{p \in \mathcal{Q}} M_{p,y} I_{p,y}} \tag{7}$$

is then the stablecoin share those pairs deliver in year y. Write $\mathcal{N}$ for every ordered pair, $\mathcal{N}_M \subseteq \mathcal{N}$ for the pairs that traded in both years, and $\mathcal{N}_V \subseteq \mathcal{N}_M$ for the pairs that carried an intermediary in both years, so that $S_y = F_y(\mathcal{N})$ and $\mathcal{N}_V$ is the continuing population of Equation (6). Bridging the aggregate to the narrower population twice and allocating what is left inside $\mathcal{N}_V$ gives

$$\Delta S = \underbrace{\Delta\big[F(\mathcal{N}) - F(\mathcal{N}_M)\big]}_{\text{markets start or stop trading}} + \underbrace{\Delta\big[F(\mathcal{N}_M) - F(\mathcal{N}_V)\big]}_{\text{vehicle role starts or stops}} + \underbrace{\phi_M + \phi_I + \phi_s}_{\text{movement inside } \mathcal{N}_V} , \tag{8}$$

where ϕ_M, ϕ_I, and ϕ_s give each factor's average marginal contribution to $F(\mathcal{N}_V)$ over the six orders in which market activity, vehicle incidence, and stablecoin share can be carried from their 2024 to their 2026 values. Averaging over orders is what lets the three exhaust $\Delta F(\mathcal{N}_V)$ without charging an interaction to whichever factor happens to move first. Read in this order, the classical vehicle-currency question of which intermediary minimises the cost of exchanging a given pair of assets

(Krugman, 1980) is the last of the five terms. The four before it record where trading happens; only the fifth records which intermediary a trade selects.

Pairs entering or leaving the sample contribute +9.8 pp; established markets that gain or lose a vehicle route, −0.4 pp; market activity shifting across the pairs that carry a vehicle in both years, +7.9 pp; and changes in how often those pairs use a vehicle at all, +7.0 pp. Direct stable-for-native substitution among them contributes +1.3 pp. The two factorisations agree on which pairs continue and on nothing else below that. The three allocated terms sum to the change in $\mathcal{N}_V$'s own stablecoin share, while the identity's within-pair and reweighting terms sum to that same change multiplied by the share of choice mass $\mathcal{N}_V$ carries, averaged over the two years. That relation holds exactly, and it is why a component of one panel is never the same object as the similarly named component of the other. Equation (8) also cuts finer. It splits the pairs traded in only one year into markets that stop or start trading altogether and established markets that keep trading and stop or start using a vehicle, and it splits the reweighting of continuing pairs into how much those pairs traded and how often they used an intermediary at all. The distance this opens is small where it matters most: the within-pair term is +1.3 pp under the finer factorisation and −0.1 pp under the identity, and both sit inside the confidence interval of the matched estimate reported in Panel D, whose upper limit is +1.7 pp. Whichever way the count total is cut, comparable trades did not switch intermediary.

The bridge's smallest term is the one that says the most once its population is opened. Pairs that gain or lose a vehicle route contribute −0.4 pp, which invites the reading that established markets barely changed whether they used an intermediary at all. Panel A partitions each year's choice mass into three classes, and the class behind that term is both narrow and closing: markets traded in both years whose vehicle role appears in only one carry 3.0% of 2024 choice mass across 4,967 ordered pairs, and 1.0% of 2026 choice mass across 1,556. Its routed activity passes through a stablecoin more often than either other class does in either year, 47.0% in 2024 and 76.9% in 2026, against 25.0% and 41.2% for the 26,547 pairs that carry a vehicle in both years and 3.9% and 43.1% for pairs whose market is observed in only one. The margin on which an established market starts or stops using an intermediary at all is therefore narrow, shrinking, and routed through the challenger where it is used. That bounds a third account of the rotation, in which a stablecoin gains aggregate share because markets that had been trading directly begin routing indirectly and use a stablecoin when they do. The account is confined to a population never larger than 3.0% of routed choice mass and cannot carry a change of +25.7 pp. Membership in the class turns over by construction: a pair belongs to it only if its vehicle role is present in one year and absent in the other, so no pair in it carries an intermediary in both years and none can register a change of intermediary. The distance between the class's two stablecoin shares records which markets occupied it in each year. Appearance in the class and disappearance from it answer to the same forces as the routing decision itself.

Conditional on the same ordered source–destination pair, calendar date, and single- or cross-exchange route type, stablecoin share changes by +0.2 pp by count, with a standard error of 0.8

pp, and by −1.3 pp by value, with a standard error of 2.2 pp. The corresponding 95% confidence intervals rule out within-pair changes larger than +1.7 pp by count and +3.0 pp by value. The matched-pair estimates are therefore small relative to the aggregate rotation.

Panel D estimates the 2024–2026 difference after absorbing variation specific to an ordered source–destination pair, calendar date within the year, and observed single- or cross-exchange route type. Let c index each such combination and let $s_{c,y}^{(m)}$ denote its stablecoin share under measure m. We estimate

$$s_{c,y}^{(m)} = \alpha_c^{(m)} + \beta^{(m)}\mathbf{1}\{y = 2026\} + \varepsilon_{c,y}^{(m)}, \qquad y \in \{2024, 2026\}. \tag{9}$$

An observation is one pair–calendar-date–route-type combination in one year. Weighted least squares uses the number or dollar value of native-plus-stable routes as weights; 2024 is the omitted year. Standard errors are clustered by ordered pair and calendar date. The comparison remains descriptive because the pairs traded and the use of one or several exchanges can respond to the same forces as intermediary choice.

Conditioning this finely has a price, and the price is coverage. Of the 762,737 ordered pairs that carry a native-or-stable route in either year, 26,547 carry one in both and make up the continuing block of Equation (6). Requiring the same calendar date and the same route scope as well leaves 5,432 pairs by count and 5,278 by value. Those pairs carry 14.0% of 2024 routed activity and 24.3% of 2026, but 41.7% and 39.3% of the dollars moved. Measured inside the continuing block alone, the same four figures are 25.2% and 50.1% by count, 50.8% and 62.9% by value. The matched estimate therefore covers a minority of route counts and a much larger part of the value, because dollars concentrate in the pairs and days observed on both sides of the comparison. Amiti et al. (2022) read the same gap between a currency's share of transactions and its share of traded value as evidence about which trades are the small ones; here it says which trades the matched comparison retains. The identity supplies the complement, since its within-pair term is taken over the whole continuing block under no date or scope condition: −0.1 pp by count and 0.0 pp by value, beside the matched +0.2 pp and −1.3 pp. Each matched interval contains the identity term of its own weighting. The narrow conditioned comparison and the wide accounting term put within-pair substitution in the same place, so its absence is not a property of the matched sample.

An entry cut gives the same reading from the first observation of a market instead of from a comparison of markets alive in both years. Among ordered pairs first observed in January through June 2026, 25.1% of first-window route mass is stablecoin-intermediated; the analogous 2024 entrant share is 5.0%. The large-notional entry margin is sharper: among entry-day pair-scope rows with at least $100,000 of within-20-percent native-or-stable supported value, the stable share rises from 48.4% to 97.8% over 16,686 2026 routes. The increase is not only a cross-venue artifact, since single-venue entrant scopes rise from 4.5% to 19.1% and cross-venue scopes from 6.8% to 37.4%. The first observed vehicle also persists. For 2026 entrants with a complete 30-day follow-up window, stable-born pairs send 97.5% of later native-plus-stable route mass through stablecoins, while native-born pairs send 0.5%; the gap is +97.0 pp with a standard error of 1.0 pp. At 120 days the corresponding gap remains +98.8 pp, with standard error 0.4 pp. This is not only an unconditional contrast. For each entry horizon $h \in \{30, 120\}$, the controlled screen estimates

$$\begin{aligned} S_{p,t+h} = \alpha_h & \\ & + \beta_h S_{p,t}^{\text{entry}} \\ & + \gamma_h \mathbf{1}\{S_{p,t}^{\text{entry}} > 1/2\} \\ & + X'_{p,t}\delta_h \\ & + \varepsilon_{p,t+h}, \end{aligned} \tag{10}$$

where $S_{p,t+h}$ is the follow-up stable-vehicle share for non-WETH entrant pair p, $S_{p,t}^{\text{entry}}$ is the first-window stable share, and X contains entry size, cohort year, endpoint class, direct-route share, and route-complexity share. Among those non-WETH entrants, a 10 percentage point higher entry-day stable share predicts +8.6 pp more stable follow-up share at 30 days and +7.4 pp at 120 days, with standard errors 0.6 pp and 1.3 pp; stable-dominant entry predicts +57.3 pp higher probability of stable-majority follow-up at 120 days, with standard error 14.8 pp. The supported split is mainly the thin-corridor case: the 120-day panel has 512,020 no-direct-route entrant rows and 181 direct-route-present rows. Inside the no-direct margin, a 10 percentage point higher entry stable share predicts +8.9 pp more stable follow-up share, with standard error 1.3 pp. The value-supported analogue is similar: a 10 percentage point higher entry stable-value share predicts +9.0 pp more follow-up stable-value share at 30 days and +7.8 pp at 120 days, with standard errors 1.0 pp and 1.4 pp, in value-weighted non-WETH entrant rows. Stable birth is also related to market continuation: among 512,201 non-WETH entrants, stable-present birth predicts +9.2% more non-entry-day primary routes over 120 days and +5.9 pp higher probability of trading again, with standard errors 1.7% and 0.7 pp. Persistence is not only a route-share average. Among 2026 stable-dominant entrants that trade again, 92.2% never register a non-stable-majority active day over the first month, and 95.1% never do so in the complete 120-day sample. The WETH endpoint identity is visible but does not exhaust the result: WETH-endpoint entrants carry 18.9% of 2026 entrant route mass, and among entrants without WETH at an endpoint stable share still rises from 0.9% to 7.7%. The secure-volume mechanism in Krugman (1980) is visible within that non-WETH margin. In 2026, entrants with a stablecoin already at an endpoint route 17.5% of their first-window mass through stablecoins, versus 5.2% for other non-WETH endpoints, and the stable-endpoint gap expands by +10.3 pp from 2024 to 2026. After entry size, endpoint class, direct-route share, and route-complexity controls, the 2026 stable-endpoint interaction is +2.4 pp, with standard error 1.0 pp. The stable-entrant margin is not a diffuse stablecoin class either: in 2026 USDC carries 73.8% and USDT 25.8% of stable-entry route mass, 99.6% together. Identity persists inside that pair: USDC-born stable entrants keep 90.5% of stable follow-up routes in USDC over 30 days and 88.3% over 120, while the corresponding USDT shares are 77.8% and 87.7%. The controlled candidate-level analogue uses 6,087 non-WETH stable-entry candidate rows with candidate fixed effects. Conditional on positive stable follow-up mass, a 10 percentage point higher entry share for a named stablecoin predicts +9.8 pp more own-candidate follow-up share at 30 days and +9.7

pp at 120 days, with standard errors 0.04 pp and 0.23 pp. In a weighted entrant regression that drops WETH-endpoint pairs, the 2026 coefficient remains +4.3 pp with standard error 0.4 pp after endpoint class, entry size, direct-route, and route-complexity controls; the additional 2026 stable-endpoint term is +5.7 pp with standard error 1.3 pp. A route-architecture screen points in the same direction: the 2026 interaction with direct-route share is +50.2 pp with standard error 13.8 pp, and the 2026 interaction with complex-route share is +35.5 pp with standard error 11.6 pp. These entry, endpoint, architecture, and identity screens are the entry analogue of the selection margin in Amiti et al. (2022): composition can move an aggregate while choices inside continuing units remain sticky. This is not evidence that the first route causes later routing or that stablecoins displace an active direct market. It says that new corridors, especially those that must use an intermediary, tend to arrive already attached to a vehicle regime, and that the regime is highly persistent over the first four months.

The same identity stickiness appears in a stress episode. During the March 2023 USDC/SVB window, stable-vehicle route activity rose by 73.6% to 8,931 routes per day, but USDC's share inside the stable-vehicle class was 67.9%, against 68.0% in the prior window, a change of −0.2 pp. USDT's share rose only afterward, to 25.0% in the post-window. The capital side diverged from the route shock: inside the five-candidate set, stablecoins' route share rose by +11.4 pp during the window while their deposited-capital share moved −2.0 pp, and after the window WETH's capital share was +3.1 pp higher than before. Stablecoin pegs depend on arbitrage and redemption arrangements (Lyons and Viswanath-Natraj, 2023; Anadu et al., 2025), so this peg shock remains observational. Even so, the episode is informative about identity under stress: vehicle identity stayed with USDC while its cash-like promise was under pressure, and LP capital stayed with the native reserve stock rather than chasing the temporary stable-route surge.

Market birth and thin-market turn-on do not mean the result is confined to one-off routes. On the same matched January–June calendar, stablecoin share among pair-days with at most five realised routes rises from 6.9% to 20.9%, with a change of +14.0 pp and standard error 1.1 pp. Among pair-days with more than 100 realised routes, the stable share is already 64.8% in 2024 and rises to 73.8% in 2026. Active pair-days carry +52.9 pp more stable share than thin pair-days in 2026. The same check at market birth is sharper. Entrants with more than 100 first-day routes carry 20.2% of 2026 entrant route mass and route 91.4% through stablecoins, up by +32.6 pp from 2024. Thin entrants in 2026 route only 6.2% through stablecoins. Because realised market size is an outcome, this screen is not a size effect; it rules out the weaker interpretation that the composition margin lives only in isolated low-activity corridors.

A second exploratory screen asks where stablecoin use appears after no baseline stable vehicle. The benchmark is sticky regimes inside continuing markets: among single-venue transition cells, 2024 stable-majority regimes remain stable-majority in 94.5% of 2026 observations, with standard error 1.8 pp, and native-majority regimes remain native-majority in 95.5%, with standard error 0.6 pp. In the count-share transition panel, a one-log-point larger baseline market is associated with a −2.7 pp lower turn-on probability, with standard error 0.4 pp, and a one-standard-deviation

Table 3: Market birth and later vehicle use

Screen	Outcome	Scaled regressor	Effect (pp)	Obs. / clusters
Entry persistence	Stable share, 30 days	Entry stable share (+10 pp)	+8.6*** (0.6)	599,870 / 333
Entry persistence	Stable share, 120 days	Entry stable share (+10 pp)	+7.4*** (1.3)	512,201 / 243
Entry persistence	Stable-majority follow-up, 120 days	Stable-majority entry	+57.3*** (14.8)	512,201 / 243
Thin-corridor split	Stable share, 120 days	Entry stable share (+10 pp), no direct route	+8.9*** (1.3)	512,020 / 243
Value-weighted entry	Stable value share, 120 days	Entry stable value share (+10 pp)	+7.8*** (1.4)	427,625 / 243
Named-stable identity	Own-stable follow-up share, 120 days	Entry named-stable share (+10 pp)	+9.7*** (0.2)	6,087 / 243
Secure-volume endpoint	Entry stable share	Stable endpoint × 2026	+5.7*** (1.3)	615,333 / 363
Route architecture	Entry stable share	Direct-route share × 2026 (+10 pp)	+5.0*** (1.4)	615,333 / 363
Route architecture	Entry stable share	Complex-route share × 2026 (+10 pp)	+3.6*** (1.2)	615,333 / 363

Note: Each cell reports a scaled regression effect in percentage points, with entry-date-clustered CR1 standard errors in parentheses. Entry-share rows scale the regressor by 10 percentage points; stable-majority entry and stable-endpoint rows are binary effects; direct-route and complex-route rows are 2026 interactions scaled by a 10 percentage point increase in the route-share regressor. The unit is a non-WETH entrant pair-scope row except for the named-stable identity row, which is a stable-candidate entry row with candidate fixed effects. Rows are route-weighted except the stable-value row, which uses within-20-percent supported value. Stars denote 1%, 5%, and 10% significance.

increase gives −3.7 pp; stable-majority switches move the same way, −1.1 pp with standard error 0.2 pp. Baseline direct-route and complex-route shares enter positively, +27.1 pp with standard error 6.4 pp and +20.3 pp with standard error 5.1 pp. The direct-route association is stronger in thinner markets: the direct-share by one-log-point-thinner interaction is +3.1 pp, with standard error 0.9 pp.

A daily native-only hazard gives the timing version of the same margin. Among 1,690,478 active pair-days with no stable vehicle at the origin date, the probability that a stable vehicle appears over the next 30 days is 24.5% in the highest market-thickness bin and 2.0% in the lowest, a gap of +22.5 pp. The oldest native-only corridors show 41.4% turn-on against 1.6% for the youngest, a gap of +39.9 pp. With month-day fixed effects, ordered-pair and date clustering, and market-route weights, one log point of origin market routes predicts +3.5 pp higher 30-day turn-on probability, with standard error 0.7 pp; pair age predicts +1.6 pp, with standard error 0.2 pp. Stable vehicles appear after a corridor has repeatedly produced route demand, but they are not simply taking native share inside an already mixed same-day set.

A mixed risk-set check makes the last point directly. Among observed pair-day-scope sets in which native and stable candidates both appear and at least five routes trade, stable candidates carry 28.3% of 2026 routes, up from 24.3% in 2024. With pair-day-scope fixed effects, however, a stable row carries −51.9 pp less route share than a native row, with standard error 6.6 pp; the 2026 interaction is +3.7 pp, with standard error 13.2 pp. Candidate network reach explains more of the within-set allocation. Adding the candidate's same-day pair-scope count outside the current risk set, one log point of reach predicts +9.4 pp more route share, with standard error 2.9 pp, while the stable-row coefficient remains −21.5 pp and the 2026 stable interaction is +2.3 pp with standard error 10.6 pp. Splitting stable rows by issuer sharpens the boundary: after the same risk-set effects and reach control, the generic stable-2026 term is −40.9 pp, while the USDC and USDT 2026 issuer interactions are +53.0 pp and +88.2 pp, with standard errors 19.3 pp, 22.5 pp, and 24.2 pp. The result is not only contemporaneous: replacing same-day reach with the candidate's pair-scope reach over the prior 30 calendar days gives +7.8 pp, with standard error 2.5 pp, and the stable-row coefficient remains negative at −20.4 pp. These are driver screens, not treatment designs. Together they place formation at the edge of incumbent regimes and in corridors with repeated demand, while pointing away from simple same-risk-set conquest.

The joint condition is easiest to read in the estimator's own unit. An observation is one pair–calendar-date–route-type cell, and 94,260 cells clear the condition, out of the 1,726,215 that carried a native-or-stable route in 2024 and the 815,483 that carried one in 2026, or 5.5% and 11.6% of the two years' cells. Those cells carry 14.0% and 24.3% of the two years' routes, so a matched cell is 2.6 times as busy as the average active cell in 2024 and 2.1 times as busy in 2026. On the dollar-weighted perimeter, where the value-agreement requirement empties 142,972 cell-years that the count measure keeps, 5.7% and 11.6% of cells are matched, and a matched cell carries 7.4 times the average cell's dollars in 2024 and 3.4 times in 2026. A cell survives only when the same market traded on the same day of the calendar year in both years, and the chance of that

Table 4: Decomposition of the stablecoin rotation

Margin or estimate	Estimate in pp	Obs.
Panel A. Route-count share: market activity and vehicle incidence		
Pairs entering or leaving the sample	+9.8 pp	
Pairs gaining or losing a vehicle route	−0.4 pp	
Market activity shifting across continuing pairs	+7.9 pp	
Change in how often continuing pairs use a vehicle	+7.0 pp	
Stablecoin share within continuing vehicle-using pairs	+1.3 pp	
Total route-count change	+25.7 pp	
Panel B. Route-count share: continuing and year-specific pairs		
Stablecoin share within continuing pairs	−0.1 pp	
Vehicle activity shifting across continuing pairs	+8.6 pp	
Weight of continuing versus year-specific pairs	−0.5 pp	
Pairs traded in only one year	+17.8 pp	
Total route-count change	+25.7 pp	
Panel C. Dollar-weighted share: continuing and year-specific pairs		
Stablecoin share within continuing pairs	0.0 pp	
Vehicle activity shifting across continuing pairs	+26.2 pp	
Weight of continuing versus year-specific pairs	−2.5 pp	
Pairs traded in only one year	+19.2 pp	
Total change in dollar-weighted share	+42.8 pp	
Panel D. Matched ordered-pair estimates		
All two-leg routes, count share	+0.22 (0.76)	188,520
20% agreement sample, count share	+0.32 (0.75)	182,834
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,834

Note: Panel A decomposes the route-count share change between 2024 and 2026 over observed market activity, vehicle incidence, and stablecoin share; common-role pairs use a native or stablecoin intermediary in both years, and a Shapley allocation distributes interactions across the five reported components. Panels B and C report the four terms of Equation (6), which conditions on native-plus-stable choice mass throughout; Panel C covers dollar-weighted routes whose source, intermediary, and destination values agree within 20%. Panels A and B decompose the same route-count total by different factorisations, so their components are not comparable term by term. Panel D estimates Equation (9); standard errors use two-way clustering by ordered pair and calendar date.

rises with how often the market trades. The matched estimate is therefore a statement about the part of the market that trades routinely, and the dollar multiple says that this part is where value concentrates. It speaks less for markets that trade intermittently, whose cells drop out even when the pair itself trades in both years. That division suits the question: the rotation is a fact about where dollars route, and the matched comparison is measured on the cells that carry two-fifths of them in either year.

Individual pairs make the accounting concrete. Among pairs whose two assets carry a recognised ticker, the largest contribution to the reweighting term comes from USDT → WETH, which is stablecoin-intermediated in both years and grows from 7,447 to 54,112 routes, or from 0.36% to 4.52% of routed activity, and adds +2.17 pp on its own. The largest such contribution among pairs absent from 2024 comes from USDS → WETH, which supplies 3,390 stablecoin-intermediated routes in 2026 and adds +0.13 pp. The largest such contribution within a continuing pair, from USDC $\rightarrow$ WBTC, adds +0.05 pp as its stablecoin share moves from 10.5% to 35.9%; the within-pair term summed over all pairs is −0.1 pp. Because most newly traded assets have no such ticker, each named pair understates its margin, which is why the margin totals accompany it. New listings or changes in pool access can bring a pair into the sample; trading activity can shift across continuing pairs; and direct-market depth or router access can change the use of indirect execution. Through these margins, routed activity reallocates across source–destination pairs. This reallocation carries most of the stablecoin increase. Stable-for-native substitution within continuing vehicle-using pairs contributes far less.

A margin of a given size can come from one corridor or from thousands of small markets, and the totals do not distinguish the two. It is therefore natural to ask whether a few large markets carry the reallocation. The two composition margins answer differently. Pairs traded only in 2026 contribute through 18,446 ordered pairs, the largest of which supplies 3.2% of that margin, and 1,386 of them are needed for nine-tenths of it. Reweighting among continuing pairs is top-heavy instead: 7 pairs supply half of its +17.2 pp gross gain, and the margin nets to +8.6 pp only because 2,817 continuing pairs lose activity weight, −8.6 pp between them. Dollar weighting sharpens the same contrast, with 5 pairs behind half of a +38.0 pp gross gain. Within continuing pairs, intermediary choice moves in both directions: 1,575 of them raise their stablecoin share and 1,489 lower it. The pairs that rise contribute +1.3 pp by count and +2.3 pp by value; the pairs that fall give back −1.4 pp and −2.4 pp. The near-zero net term is an offset between markets moving in both directions, not an absence of movement, and it stays small beside the weight that moves across pairs. The rotation therefore travels along two different edges of the trading network: the network extends into many new and mostly small markets, while activity weight concentrates in a few corridors that were already large. This reorganisation of where trading happens carries the aggregate change; the growth of a single corridor does not. The two edges also separate what a currency network measures: a matrix recording which pairs trade at all, as in the historical exchange structure of Flandreau and Jobst (2009), registers the first margin and is silent on the second.

The vehicle role is not contestable everywhere in the pair space. A route whose source or destination is wrapped ether cannot also use wrapped ether as its intermediary, so within the native-versus-stablecoin comparison such a route is stablecoin-intermediated by definition, in 2024 and in 2026 alike. Since the pairs that grew fastest are concentrated among exactly those corridors, a question of interest is whether the two composition margins record anything more than the growth of markets whose vehicle was never in doubt. We split each margin according to whether wrapped ether is an endpoint of the ordered pair. The split is decided by one contract address rather than by a ticker, so the long tail of newly traded assets is classified as reliably as the majors. 1,499 of the 26,547 continuing pairs, or 5.6%, have such an endpoint, and their share of routed activity rises from 20.9% to 33.0%. By route count they supply +6.3 pp of the +8.6 pp reweighting margin and +13.9 pp of the +21.0 pp new-pair margin, so most of the count rotation does run through corridors that had no intermediary to choose. Dollar weighting reverses the reading. The same corridors supply +11.2 pp of the +26.2 pp value reweighting margin, or 42.6% of it, and +3.6 pp of the value new-pair margin, or 16.0%. Once trades are weighted by the value they move, the rotation is carried by pairs that could have been routed either way and were routed through a stablecoin. The same identity disciplines the within-pair term: a pair with a wrapped-ether endpoint contributes exactly zero to it under both weightings, so the near-zero total is not an average diluted by markets that were frozen at a stablecoin share of one. Every unit of it comes from pairs whose intermediary was a live choice, and among those pairs the change is still near zero.

So far the split has been taken over all routes at once, which does not say where in the venue structure the corridors without a vehicle choice sit. Taking it separately for routes that stay on one exchange and routes that cross exchanges answers that, and the two weightings answer it differently. Within a single exchange, reweighting among continuing pairs contributes +4.4 pp, of which wrapped-ether-endpoint corridors supply 55.3%. Across exchanges the same margin is +11.4 pp, and those corridors supply 82.9% of it. What survives the split is the part contributed by pairs that had a choice, and it is the same size in both: +2.0 pp within an exchange and +1.9 pp across exchanges. The entire cross-exchange excess in the count margin is therefore corridors that were stablecoin-intermediated by definition, and the corridors that could have gone either way reallocate activity at the same rate wherever the route runs. Weighting by the value moved removes the contrast. The margin is +26.6 pp within an exchange against +21.4 pp across exchanges, and eligible corridors supply 48.6% and 50.5% of them. In route counts, then, the rotation looks like something that happened to cross-exchange trading, and what it records there is which pairs became active across exchanges; in value terms it is no more a cross-exchange event than a single-exchange one. Makarov and Schoar (2020) decompose bitcoin price dispersion the same way and find the across-region component both the larger one and, once traded volume is taken into account, the one that carries the economic magnitude. The across-venue excess here does the opposite: it is visible in unweighted route counts and gone from the dollar-weighted margin.

That split was taken of continuing markets, whose endpoints are fixed and whose intermediary can change. The larger composition margin comes from pairs that traded only in 2026, and it is there that a reader should most expect eligibility to be the whole story: a market appearing for the first time is disproportionately a newly listed asset whose only quote is a wrapped-ether pair, which is to say a corridor that arrives with its intermediary already settled. By route count that expectation is met, and the route scope barely moves it. Entry contributes +21.4 pp within a single exchange and +23.8 pp across exchanges, and wrapped-ether-endpoint corridors supply 65.8% of the first and 67.5% of the second. Dollar weighting separates the two scopes instead of reversing them. The entry margin is +14.1 pp within an exchange and twice that, +28.5 pp, across exchanges, while the part from corridors that had no vehicle to choose is +3.4 pp and +4.0 pp, almost the same in both. What differs is the part that had a choice: +10.7 pp within an exchange and +24.5 pp across it, so eligible corridors account for 24.1% of the margin within an exchange and 14.2% across it. The entry margin is therefore the mirror image of the reweighting margin. There the cross-exchange excess was carried entirely by corridors whose vehicle was never in question; here it is carried entirely by corridors that had one and were routed through a stablecoin anyway. Read together, the two say that a market already trading across exchanges gains stablecoin share mainly where no other intermediary was available, while the value arriving with markets that begin to trade across exchanges comes from corridors that could have been routed through the native token.

Those splits sort corridors by whether the vehicle was fixed and by where the route ran. Neither describes the corridors that had a choice. We divide them once more, on whether a stablecoin stands at an endpoint of the ordered pair. A second eligibility rule in disguise would carry no information, and this is not one: a trade into one dollar token can be carried by wrapped ether or by a different dollar token. The 10,278 continuing corridors of this class route through a stablecoin in 5.6% of their 2024 activity and 12.8% of their 2026 activity, well below the unit share eligibility imposes, and they hold 68.8% and 60.5% of routed activity in the two years. By route count they supply 26.0% of the reweighting margin and 28.7% of the entry margin, less in both cases than wrapped-ether corridors supply. Dollar weighting puts them ahead. The same corridors carry +15.0 pp of the +26.2 pp value reweighting margin, or 57.3% of it, and +16.9 pp of the value entry margin, or 76.4%, both larger than the eligible corridors' 42.6% and 16.0%. Corridors holding neither candidate at an endpoint are the many and the small: 142,831 of them begin trading in 2026 and 13,390 trade in both years, and they supply +1.7 pp and +0.0 pp of the two value margins. Inside the dollar-endpoint corridors the movement is once again reallocation. Their activity-weighted stablecoin share by value rises from 21.0% to 61.6% while their within-pair term is −0.0 pp, so the rise is weight moving toward corridors that already routed through a dollar. Weighted by the value they move, the markets that carry the rotation are markets with a dollar already at one end, and the intermediary inside each of them barely moved on net. Krugman (1980) calls the trading a currency draws from payments that never pass through it its secure volume, and holds that a currency short of it cannot take the vehicle role from an incumbent however transaction costs fall. The corridors carrying this rotation are the ones where the challenger's secure volume sits.

That entry margin is gross. The decomposition subtracts the corridors that stopped trading, and the netting is not an accounting nuisance: it is the comparison the term makes. Averaged across the two years, corridors observed in only one of them carry 48.0% of route activity and 27.8% of the value moved, and the netted term is that mass multiplied by the gap between how two populations were routed—the corridors that left and the corridors that arrived. The two are not alike. The departing corridors sent 6.8% of their routes and 10.8% of their dollars through a stablecoin; the corridors that replaced them send 43.8% and 79.8%. Removing the wrapped-ether-endpoint corridors, whose vehicle was settled before they opened, leaves the departing corridors at 1.7% by route count and 3.2% by value, and the arriving ones at 21.0% and 76.9%. Across exchanges the arriving corridors that could have chosen anything route 88.3% of their dollars through a stablecoin. The largest margin in the count decomposition is therefore not adoption inside a settled set of markets. It is the replacement of one population of markets by another that routes differently from the day it opens. Flandreau and Jobst (2009) estimate the incumbency advantage of an international currency across a stable roster of trading relationships and find persistence without lock-in; here the roster is what turns over, and the challenger's gain is concentrated in the corridors that did not exist when the incumbent's network was formed.

Two different economies produce that gap, and the two weightings do not agree on which one is at work. Corridors are of three kinds: those with wrapped ether at an endpoint, which cannot use it to intermediate and route through a stablecoin by construction; those with a stablecoin at an endpoint, which may still be carried either way; and those with some other asset at each end. Evaluated at the two cohorts' midpoints, the gap separates exactly into the mix of kinds each cohort brings and the rate at which each kind routes. By route count the mix carries 65.9% of the +17.8 pp netted term, and nearly all of that is the first kind: wrapped-ether-paired corridors are 28.9% of arriving activity against 5.2% of departing activity, worth +11.4 pp by themselves. They enter the margin through their weight alone, because their routing rate is one in both cohorts. The remaining +6.1 pp is a rate difference among corridors that could have been carried either way. By value the ordering reverses: the mix contributes +3.0 pp and the rates +16.2 pp, or 84.6% of +19.2 pp. Arriving corridors with a stablecoin at an endpoint send 81.7% of their dollars through a stablecoin where the departing corridors of the same kind sent 6.3%, and arriving corridors with some other asset at each end send 48.3% against 0.9%. No corridor appears on both sides of those comparisons, so none of these figures describes an intermediary being changed; the dollars now moving through corridors that opened after the challenger was established were never routed the old way. Dowd and Greenaway (1993) make an agent's willingness to switch depend on the expectation that others will switch too, and the excess inertia that follows is a claim about the agents already holding the incumbent. Their model is silent on the agents who arrive later, and it is those the data separate: inside a continuing corridor the intermediary barely moves, while the corridors that open once the challenger is available carry it from their first trade.

Replacement on that scale moves activity between the two populations as well as changing how each is routed, and the decomposition prices the two separately. Corridors observed in both years fell from 55.5% of route activity to 48.6%, a shift of −6.9 pp, and from 82.0% of the value moved to 62.4%, a shift of −19.6 pp; across exchanges the dollar shift is −27.6 pp. Roughly a fifth of the value in the comparison therefore changed which population of corridors carried it. Against those magnitudes the term itself is −0.5 pp by route count and −2.5 pp by value, the smallest of the four and the only one no other margin overlaps. It is small not because little activity moved but because the two populations are priced almost alike: averaged over the two years, corridors alive in both route 33.1% of their routes and 58.2% of their dollars through a stablecoin, against 25.3% and 45.3% for corridors alive in only one, gaps of +7.8 pp and +12.9 pp. The term is negative in every weighting and every route scope: holding both populations at those midpoint routing rates, the activity that changed population lowered the stablecoin share rather than raising it. The rotation is thus not activity migrating into corridors that already used the challenger. It is that the population receiving the activity routes differently from the one it replaced. Amiti et al.

(2022) find invoicing currency to be close to a fixed attribute of an exporter, switching for a small minority of observations over almost four years, with the composition of destinations explaining little of the cross-sectional variation. Choice is similarly sticky inside a continuing corridor here; the difference is that the population of corridors turns over fast enough for composition to carry the whole aggregate change.

Composition is measured on a population that turns over, and the same measurement can be taken where it holds still. Fix a set of corridors and the identity's four terms collapse to the two that operate inside it. The two readings of the rotation then predict opposite outcomes. Under turnover, holding the set fixed removes the rotation. Under reallocation among corridors present throughout, holding the set fixed leaves it, smaller by whatever the entrants supplied. A cohort is fixed here when its cells carry at least one observed route in both endpoint years. One cohort serves both weightings, and the episode and dollar estimates therefore restrict the same units.

Over 543 matched days on the exact two-leg sample, the pooled change is +25.42 pp by route count and +43.87 pp by value. Restricting to the 39,748 corridors active in both years leaves +14.85 pp, with a standard error of 1.02 pp, and +39.36 pp, with a standard error of 2.18 pp. Those corridors carry 55.5% and 57.7% of the two years' routes and 82.2% and 77.1% of their dollars. Most of the dollar rotation and more than half of the count rotation are therefore observed inside corridors present on both sides of the comparison. The identity's within-corridor term of −0.1 pp by count, whose matched-cell interval reaches only +1.7 pp, then places the surviving rotation: activity moves across established corridors toward the ones already routing through a stablecoin.

The intermediary menu inside a corridor turns over as well, and a second cohort holds it fixed. Requiring the corridor and the intermediating candidate to appear in both endpoint years leaves 40,368 corridor–candidate cells, a change of +12.94 pp by count and +37.24 pp by value, and 80.4% and 66.1% of the two years' dollars. Candidates that intermediate a corridor in one year only thus account for a small part of the measured rotation. The weight moved onto candidates already available where they were already used.

A third cohort fixes the sequence of exchanges a route ran through, and it is reported for the bound it supplies. That sequence is selected jointly with the intermediary. Conditioning on it absorbs part of the rotation's own content, and the estimate reads as a diagnostic. Inside it the count change is −0.18 pp, with a 95% interval of −2.13 pp to +1.76 pp, against a pooled +25.42 pp; the value change is +21.74 pp. Episode-weighted rotation is thus bounded to a small fraction of its pooled size once the realised exchange architecture is held constant, while dollar-weighted rotation keeps about half of its own. The cohort retains 37.3% of 2026 dollars against 77.1% for the corridor cohort, and coverage explains part of that gap. Amiti et al. (2022) separate the selection of firms into a currency from the effect of the currency they chose by entering proxies for the choice alongside the choice itself, and read which of the two dominates. The exchange sequence plays the proxy's role here, and it dominates the episode measure and shares the dollar measure. Three conditions this comparison would also impose stay outside it: the pools reachable when the route was assembled, the size of the trade, and the best alternative within reach. Each needs pre-transaction state that the route measurement here does not carry, and each is recorded as unfitted in the exhibit's support ledger.

3.3 USDT's intermediary role

USDC and USDT together account for 92.1% of the increase in stablecoin count share. We examine whether USDT's changing role is specific to intermediation by comparing its share of intermediary use with its share of appearances at observed route endpoints. Direct routes contribute to the endpoint denominator; longer routes contribute once for each intermediary they use. A ratio above one means that the currency appears more often as an intermediary than at either end of an observed pool route.

On matched January–June dates, the difference between USDT's intermediary and route-endpoint shares rises from −7.13 pp to +8.14 pp. The change is +15.27 pp, with a standard error of 1.50 pp. The ratio of the same two roles, shown as a descriptive full-period comparison in Table 5, rises from 1.06 in 2024 to 1.23 in January–June 2026 by count; the dollar-weighted ratio moves from 0.59 to 1.42. The matched-window gap is the basis for the transition claim.

The expansion of cross-exchange routes accounts for 6.3% of USDT's count-share increase and 14.4% of its dollar-weighted increase. The remaining 93.7% by count and 85.6% by value occurs within the single- and cross-exchange groups. This decomposition conditions on the route that was executed; it does not reveal every alternative available to the router.

The rise in USDT's intermediary use relative to its observed route-endpoint use could reflect issuance, redemption access, venue coverage, or the distribution of liquidity across pools, and some of these variables may themselves respond to routed activity. Section 5 develops tests that can distinguish these explanations.

Table 5: USDT intermediary use relative to observed route-endpoint use

	2024	2026
Count excess-use ratio (2024 full year; 2026 January–June)	1.06	1.23
Value-weighted excess-use ratio (2024 full year; 2026 January–June)	0.59	1.42
Paired January–June intermediary minus route-endpoint share	−7.13 pp	+8.14 pp

Note: Intermediary and observed route-endpoint shares are formed across native, staked-native, stable, and imported currencies on complete routes that do not return to their starting asset. Direct routes enter the route-endpoint denominator. Excess use is intermediary share divided by route-endpoint share; the gap is their difference. Ratios use the full 2024 calendar year and January–June 2026, while the share gap uses matched January–June dates. Dollar-weighted measures require source, intermediary, and destination values to agree within 20%.

3.4 The cross-section within a day

Every comparison so far reads across years, which invites the reading that the calendar is doing the work. If the vehicle role is simply whatever the sample period happens to do, then the rotation is a description of 2020 through 2026 and not of how an intermediary is chosen. The construct itself does not require the calendar. Excess use compares two roles of the same asset on the same day, so it can be asked entirely within a day, of the assets that were available that day, with the year absorbed into a fixed effect.

We therefore write Equation (5) as a coefficient. The outcome is a currency's share of the day's intermediary episodes in percentage points; the regressor of interest is its own share of that day's route-endpoint episodes in the same units; and each specification above the pooled baseline absorbs a date fixed effect, so the estimate comes from the cross-section of currencies observed on one day. This adds two quantities the ratio cannot express. The slope on endpoint share says whether intermediary use rises more or less than one for one with an asset's own trade demand, and the class intercepts say how much of the role an asset class takes beyond what its demand implies. Amiti et al. (2022) identify currency choice the same way, absorbing the macroeconomic environment in fixed effects so that what remains is the differential response of firms facing the same conditions. The ratio remains the primary extent measure; the regression is the same comparison with the calendar demoted from the identifying variation to a control. The sample is the 1,828,862 currency-days on which 304,572 currencies were quoted at a route endpoint with at least twenty route units, spanning 2,259 days. Inference clusters on date and currency, and on the few-currency cuts on date alone with a thirty-day Bartlett lag, since there the repeated sampling unit is the day.

Absorbing the calendar changes almost nothing. The native premium moves from +34.56 pp pooled to +34.55 pp within day, and the stablecoin premium from +2.42 pp to +2.42 pp; both are estimated imprecisely across currencies, with intervals of $[-10.62, +79.75]$ and $[-0.17, +5.01]$. The differences between asset classes are cross-sectional, not a period effect. What does move them is conditioning on the currency's own trade demand. The native intercept falls to +0.05 pp with a standard error of 0.56 pp, while the stablecoin intercept holds at +0.85 pp (0.50 pp). The apparent intermediary dominance of the native asset is accounted for by its own demand; the stablecoin's is not. That is the comparison the search-theoretic account of a medium of exchange asks for, since in Kiyotaki and Wright (1989) an asset is used to intermediate because of what the other assets available at that moment can do, not because of its own turnover. The difference between the two, +0.79 pp with an interval of $[-1.20, +2.78]$, is not one this cross-section can separate from zero.

The pass-through of demand into the intermediary role is 1.587 across currencies within a day (0.026), and 0.985 (0.301, $p = 0.001$) within a currency and within a day once 306,830 currency and day effects are absorbed. A currency that gains a point of trade demand gains about a point of intermediary use; the excess above one in the cross-section is composition between currencies. The slope also differs by class, but not everywhere. Native pass-through exceeds imported by 0.075, with an interval of $[+0.02, +0.13]$; the stablecoin slope exceeds the native slope by 0.351, with an interval of $[-0.17, +0.87]$ that the data cannot separate from zero. We report the second because it is the comparison the rotation would predict, and it does not hold.

Restricting the cross-section to the five currencies that carry the vehicle role sharpens it. With the stablecoin class as the base, the native asset intermediates -17.45 pp relative to a stablecoin on the same day and at the same trade demand, with an interval of $[-32.41, -2.49]$ over 11,233 currency-days. Over all 37 classified currencies the same contrast is -1.51 pp, with an interval of $[-4.09, +1.06]$. The two magnitudes answer one question at two scales, and quoting either alone would misstate its size: the candidate set concentrates the day's share mass in five currencies, so its coefficients are large, and the classified set spreads the same question across a long tail whose currencies hold almost none of it, so its coefficients are small and the contrast is not separable there. Inside the stablecoin class the role is not uniform either. Against a fiat-reserve claim on the same day and at the same demand, an on-chain-collateralised stablecoin intermediates -0.77 pp ($[-1.35, -0.20]$) and a synthetic one -0.61 pp ($[-1.11, -0.11]$), while USDT intermediates -5.32 pp relative to USDC ($[-7.14, -3.51]$). The class that gains the role over the sample is not homogeneous in it.

Three features of the design have to be measured before the estimates can be read economically. First, both variables are shares of the same day's route universe, and a currency cannot intermediate a route it terminates, so a mechanical crowd-out channel exists; it is negligible here, since the mean currency-day uses 0.0008 of the routes it does not itself terminate and only 0.08% of currency-days exceed half of that ceiling, and in any case crowd-out can only push the slope down. Second, each currency's own episodes enter the denominator of both of its shares. Recomputing both against the day's episodes excluding the currency's own raises the slope to 3.209 (0.090), so the association is not the shared denominator. Third, the twenty-unit floor is a choice, and the slope is 1.579 at five units and 1.595 at one hundred. Calendar time returns here only as a sample split: refitting on each half of the sample gives slopes of 1.605 and 1.554, and native intercepts of $+0.70$ pp $[-1.22, +2.63]$ and -0.72 pp $[-2.56, +1.12]$, both intervals covering zero. One result does not survive reweighting. Weighting currency-days by route units turns the native intercept to -15.91 pp, so its sign depends on whether each currency-day or each route counts, and we read it as descriptive.

Table 6: Intermediary role within a day

Specification	Class premium (pp) Native	Class premium (pp) Stable	Demand pass-through	Absorbed effects
Pooled	+34.56	+2.42	—	0
	(23.04)	(1.32)	—	
+ day effects	+34.55	+2.42	—	2,259
	(23.04)	(1.32)	—	
+ own endpoint share	+0.05	+0.85	1.587	2,259
	(0.56)	(0.50)	(0.026)	
+ currency effects	—	—	0.985	306,830
	—	—	(0.301)	

Note: Currency-day observations on which the currency is quoted at a route endpoint and the day supplies at least twenty route units: 1,828,862 currency-days over 304,572 currencies and 2,259 days. The outcome is the currency's share of the day's intermediary episodes in percentage points; the demand regressor is its share of the day's route-endpoint episodes in the same units. Class premiums are measured against the residual unclassified bucket, with standard errors in parentheses clustered on date and currency. The final row absorbs currency and day effects jointly, so class premiums are not identified there.

3.5 Venue span and route complexity

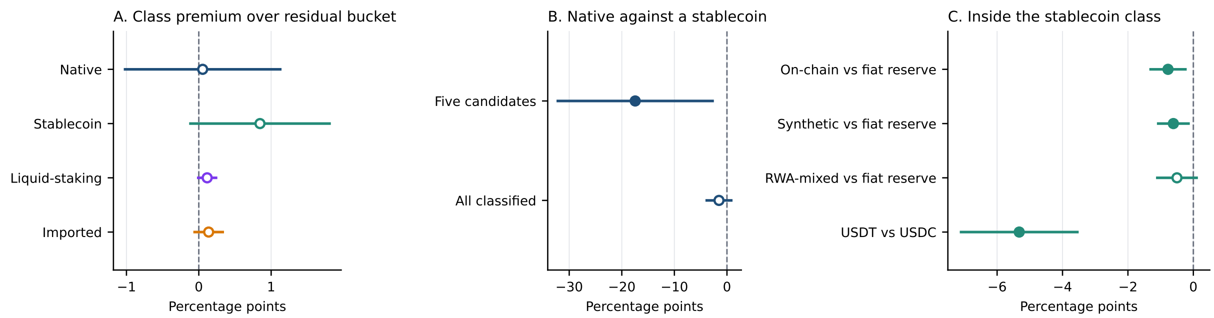
Cross-exchange routing expands sharply during the sample. Access to pools on several exchanges can favour an intermediary with liquidity dispersed across venues, while also making direct execution easier to find. Cross-exchange routes grow from 2.0% of route count in 2020 to 57.2% in 2026 and account for 79.1% of dollar-weighted route value in 2026.

Intermediation becomes less common over 2022–2026 even as cross-exchange routing grows. Its count share falls from 13.95% to 11.92% across the full exchange set and to 8.60% among five exchanges observed throughout the comparison. The changes are −2.03 pp and −5.35 pp, with standard errors of 0.39 pp and 0.38 pp. Stablecoin share nevertheless rises within every observed group defined by exchange span and route complexity. Its count-share increase is +7.45 pp larger across exchanges than within one exchange, with a standard error of 1.33 pp; the dollar-weighted estimates rise in both groups.

Those estimates compare two endpoint years. The calendar supplies the variation that identifies them. A second reading of the same stratification removes that reliance. Each trading day contributes one observation for its single-exchange routes and one for its cross-exchange routes, and a day effect absorbs everything the two halves share, from gas prices to market direction. The remaining difference measures how far apart the two halves of one day's route universe stand.

That difference is large in dollars and indistinct in counts. Over 2,234 days on which both halves are measured, one-intermediary routes that cross exchanges carry a stablecoin share +14.18 pp above routes executed on one exchange, with a standard error of 3.69 pp and a 95% interval of $[+7.0, +21.4]$. Weighting each day–group cell by the dollars behind it returns +12.12 pp with a standard error of 1.78 pp. Counting routes rather than dollars gives +3.73 pp, whose interval $[-2.2, +9.7]$ spans zero and admits a gap in either direction. Exchange span therefore tracks which

Figure 2: Within-day intermediary role, conditional on own trade demand



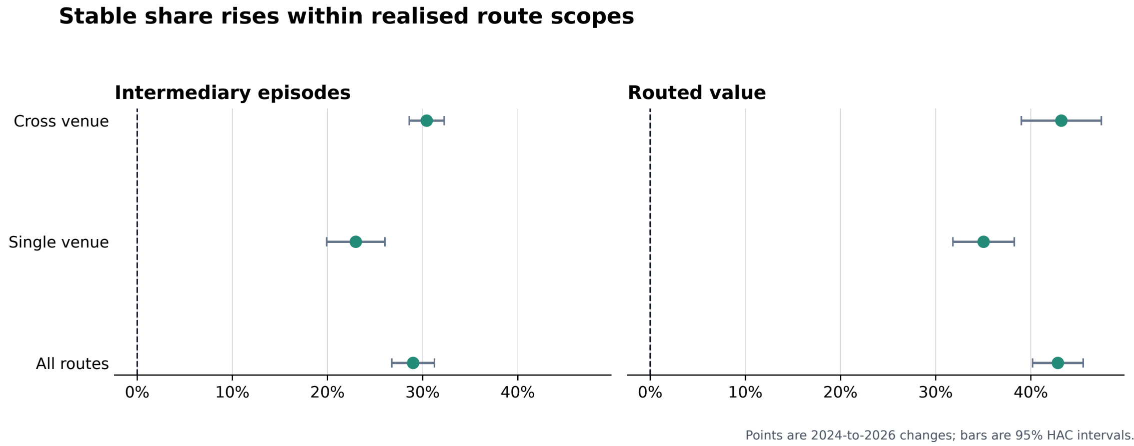
Note: Each row is one coefficient from a regression of a currency's share of the day's intermediary episodes on its own share of that day's route-endpoint episodes, both in percentage points, with date fixed effects absorbing the calendar. Bars are 95% intervals. A filled marker means the interval clears zero and an open marker means it does not; rows are drawn at the same weight either way. Panel A measures each class against the residual unclassified bucket over every endpoint-supported currency-day. Panel B measures the native asset against a stablecoin on the five named candidates and on all classified currencies; the two samples put different amounts of the day's share mass behind the same indicator, so the magnitudes are not comparable. Panel C measures backing regimes against a fiat-reserve claim and USDT against USDC inside the stablecoin class. The three panels use separate horizontal scales.

currency carries the value that moves through an intermediary, and the same day's evidence does not establish that it tracks how often an indirect route reaches for a stablecoin at all. Chen and Duffie (2021) model trade split across exchanges as thinning depth at each one while improving allocation overall. Thin depth at a single venue binds first on the largest orders, and a difference that opens in routed dollars while route counts stay level fits that reading.

The dollar gap is also unsteady across the sample. Estimated year by year, it opens at +69.7 pp in 2020, closes to −3.2 pp in 2023 on an interval of $[-9.2, +2.8]$, and reopens to +14.5 pp in 2026. A design that reads only the endpoint years of the sample would record a gap at both ends and miss the closure between them. Routes are not assigned to an exchange span, and these differences therefore describe the composition of two selected populations rather than the effect of widening a route's venue set.

Taken together, the estimates separate cross-exchange trading, the incidence of intermediation, and the composition of intermediaries. Wider venue span coincides with less intermediation overall, while stablecoins gain among the indirect routes that remain in both single- and cross-exchange groups. Section 5.1 turns that stratification into a test of whether the aggregate rotation is a by-product of how routes were built. Isolating the effect of router access requires comparisons among the same ordered source–destination asset pairs, trade sizes, and feasible paths.

Figure 3: Stablecoin intermediary share rises within single- and cross-exchange routes



Note: The sample contains complete routes that do not return to their starting asset, including routes with more than two legs. Each route contributes once for every intermediary it uses. Stablecoin shares use all intermediary types as the denominator. Count and dollar-weighted estimates are reported separately on matched January–June dates in 2024 and 2026. Dollar-weighted measures require source, intermediary, and destination values to agree within 20%.

4 Comparing direct and intermediated execution

Realised routes identify the assets that carried each trade. Execution cost is a separate object: it compares those routes with alternatives available immediately beforehand, quoted at the same trade size and over a common set of exchanges. We hold the source i, destination o, dollar input q, and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O^D_{i,o,q,e} = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p\big(i \to o, \Delta_i(q,e); \chi_{p,e^-}\big)\,, \tag{11}$$

$$O^I_{i,o,k,q,e} = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'}\left(k \to o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p\big(i \to k, \Delta_i(q,e); \chi_{p,e^-}\big)\,; \chi_{p',e^-}\right) \tag{12}$$

This comparison extends the realised-versus-optimised allocation benchmark of Xi and Moallemi (2026) from parallel pools for one asset pair to sequential routes through an intermediary.

The vehicle-route shortfall relative to direct execution is

$$\Delta C^D_{i,o,k,q,e} = \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}}, \tag{13}$$

reported in basis points as $10^4 \Delta C^D_{i,o,k,q,e}$. Positive values mean that the direct route returns more output before gas. Swaps and liquidity changes are replayed in on-chain order because a daily or hourly closing state incorporates events that occurred after the trade. The observed event marks settlement; a quote formed earlier can reflect a different opportunity set.

We compare hypothetical legs only within a common executable range. Let $\widetilde{Q}_\ell$ denote leg ℓ's output at the pool's marginal rate. Each leg satisfies

$$\pi_\ell(\Delta_i, e^-) = \frac{\widetilde{Q}_\ell(\Delta_i, e^-) - Q_\ell(\Delta_i, e^-)}{\widetilde{Q}_\ell(\Delta_i, e^-)} \leq \bar{\pi}, \qquad \bar{\pi} = 0.05. \tag{14}$$

The 5% own-price-impact ceiling applies across pool families; contract-specific limits add further restrictions. Appendix B gives the pricing equations and Appendix D calibrates the support rule.

Receipt gas provides a narrower execution-friction check. Rather than comparing pool outputs, it measures the fixed toll attached to adding a vehicle leg. That toll matters because gas fees can change both trading costs and liquidity concentration in decentralised exchange (Caparros et al., 2024). In the executed-route gas sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year's median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, but the dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. Pricing the same fixed toll at each stable-vehicle route's own gas price sharpens the interpretation: median stable-vehicle notional falls from \$832 to \$82, while the median fixed toll falls from 44.3 bp to 16.2 bp and the share of stable-vehicle routes with a fixed toll below 25 basis points rises from 39.5% to 55.7%, the trade-size version of the fixed-cost channel in Caparros et al. (2024). Thus the extra-hop overhead remains visible in gas units while becoming economically small for ordinary routed notionals. This evidence supports the interpretation that vehicle formation moved into markets where fixed execution overhead had ceased to be a large barrier; it is not an all-in proof that the vehicle route beat the direct alternative on price impact and fees.

The benchmark represents public on-chain liquidity in the declared exchange set. Private order flow, financing costs, losses from reverted transactions, and pools outside that set can alter a router's choice. Pool-family coverage is economically material. In the sampled Curve exercise, pools outside the StableSwap model carry 34.2% of assessed volume, including 65.2% of native-leg volume and 21.1% of stablecoin-leg volume. Hook-bearing or dynamic-fee Uniswap v4 pools, SushiSwap v3, and several Balancer families also use pricing states outside the benchmark. Appendix E quantifies these boundaries.

5 What can account for the rotation?

The allocation results sharpen the mechanism question. Stablecoin dominance rose chiefly as routed activity moved across source–destination pairs and the incidence of intermediation changed, while repeated replacement of a native intermediary within continuing pairs contributed little. Any explanation for the rotation must therefore account for the assets being exchanged as well as the intermediary carrying the route.

5.1 Path complexity and exchange span

Does the rotation follow from how routes are assembled rather than from which asset intermediates them? Paths grew longer and reached across more exchanges over the sample, and long paths already favoured stablecoins: in 2024 stablecoins carried 44.5% of native-plus-stable intermediary episodes on routes of more than two legs, against 16.9% on two-leg routes. An account in which construction does the work therefore predicts a specific pattern. The stablecoin share should be roughly flat inside a complexity class, and the aggregate increase should come from the movement of activity toward the longer routes that were already stablecoin-intensive.

Neither part holds. Among intermediary episodes on two-leg routes, the stablecoin share rises +25.4 pp by count, with a standard error of 1.05 pp, and +43.9 pp by value (2.02 pp). Among episodes on routes of more than two legs it rises +16.3 pp (1.86 pp) and +35.9 pp (1.68 pp), from a base of 44.5% to 60.7% by count. Both classes move, and the class with the shortest paths moves further, so reweighting toward complex routes cannot supply the aggregate change. Dividing each complexity class again by whether the route touched one exchange or several leaves four groups, and every one of them rises; the weakest is +17.0 pp by count (1.36 pp) and +25.6 pp by value (1.52 pp). The stablecoin gain is thus common to short and long paths and to routes that stay on one exchange and routes that do not.

Two boundaries limit what this comparison licenses. The number of legs records the path a trade actually took, so it measures how the route was built and not how well it was chosen; a route with more legs is not thereby a worse or a better execution. Complexity and exchange span are also outcomes rather than assignments, since the same conditions that make a longer or wider path attractive can bear on the choice of intermediary. The comparison rules out a composition account of the rotation; it does not identify an effect of routing software or of the exchange set on which asset intermediates a trade.

5.2 The software that assembles the path

Traders on these exchanges seldom assemble a path by hand. A router contract holds the pool set, searches it, and returns a sequence of legs, and the routers in widest use over this sample were rewritten three times: the Auto Router in September 2021, its cross-version successor that December, and the Universal Router in November 2022.[6] Each release widened the set of paths an ordinary trader could reach without extra effort. The alternative this raises is that the intermediary is a residue of the search: the software found a path through a stablecoin because such paths existed and had become cheap to look for, so the rotation would record an improvement in search and not a change in what traders were willing to hold between legs. Search does leave room to move. On parallel pools for a single asset pair, realised routes fall measurably short of the optimum available in the same pool state (Xi and Moallemi, 2026), so a router that closes part of that gap changes

[6] Uniswap Labs, *Introducing Auto Router*, September 16, 2021, and *Introducing Permit2 and Universal Router*, November 17, 2022. The cross-version release is dated from Uniswap Labs' v2-versus-v3 liquidity-provider study of December 16, 2021.

which paths trades actually take.

Testing this as market microstructure would test a trading technology is not available here. That literature identifies the effect of an exchange technology when the exchange grants it to some securities before others, because the securities still waiting absorb whatever else is moving in the market at the same time. A router release admits no such comparison. The contract is deployed once and is available immediately for every pair, at every exchange the router indexes, to every trader, so there is no untreated group and no difference to take. What a symmetric window around each release can measure is whether the structure of routes moved at all when the search over them was rewritten, and that weaker object is the one the alternative needs. If automated search produced the intermediary, the incidence of intermediation should step up when the search was automated.

It does not. Table 7 reports route composition over the 60 days on either side of each release, excluding the release date itself, on the daily calendar used throughout the paper. The share of economic routes carrying a third asset between the endpoints falls by −5.7 pp at the first release, rises by +0.8 pp at the second, and falls by −0.7 pp at the third, moving from 20.8% before the first release to 12.5% after the last. The largest increase at any of the three is +0.8 pp. Path length is as still: the mean number of legs on an indirect route moves by at most 0.049 at any release, and the share of indirect routes running past two legs changes by a few percentage points in either direction. Indirect routing as a whole behaves the same way, falling sharply at the first release and moving by less than half a percentage point at the other two.

One margin does move, and it is the margin a router should move. Among intermediated routes, the share touching more than one exchange rises by +2.5 pp at the first release from a base of 8.5% and by +3.1 pp at the second; at the third it changes by −2.0 pp, from a level near a quarter of intermediated routes to 24.4%. Automated search integrated the exchange set without sending trades through more assets. The distinction is the one that governs this paper's measure, which counts the asset standing between the endpoints and not the number of venues its legs visit, and it is the same separation the exchange-span stratification of Section 5.1 makes within a single date.

Three limits bound what the windows license. The dates are not independent of one another: the later window of the first release and the earlier window of the second share thirty days, so the first two comparisons are drawn partly from the same trading. Nor are they clean of the market. The earlier window of the third release contains the November 2022 failure of a large centralised exchange, and the composition change recorded there belongs at least as much to that event as to the router. A release date, finally, marks when a contract became callable and not when traders began routing through it, so a window straddling gradual adoption understates a movement that arrived slowly. What survives these limits is a narrow negative statement, and it is the one the alternative requires: at the three dates when path search was rewritten, the incidence of intermediation did not step up, while the exchange set the search reached did widen.

Table 7: Route structure around public router releases

	Indirect	Intermediated	Cross-exchange	Legs	Exchanges	Over two legs
Panel A: Auto Router, September 16, 2021 (60 days each side)						
Before	22.7	20.8	8.5	2.08	1.10	6.6
After	18.9	15.1	11.0	2.10	1.11	8.1
Change	−3.7	−5.7	+2.5	+0.01	+0.01	+1.5
Panel B: Cross-version Auto Router, December 16, 2021 (60 days each side)						
Before	18.5	13.9	11.3	2.11	1.11	8.4
After	18.6	14.8	14.4	2.16	1.16	11.3
Change	+0.2	+0.8	+3.1	+0.05	+0.05	+2.9
Panel C: Universal Router, November 17, 2022 (60 days each side)						
Before	19.8	13.2	26.4	2.19	1.22	13.1
After	19.5	12.5	24.4	2.20	1.20	12.0
Change	−0.3	−0.7	−2.0	+0.01	−0.03	−1.1

Note: Symmetric windows of 60 calendar days either side of each release, excluding the release date. Indirect is the share of economic routes with more than one leg; Intermediated is the share carrying a third asset between the endpoints; Cross-exchange is the share of intermediated routes touching more than one exchange; Over two legs is the share of indirect routes with more than two legs; Legs and Exchanges are means per indirect route. Shares and their changes are in percentage points. Economic routes exclude canonical endpoint round trips. The windows are market-wide and carry no split by intermediary currency. A release is available to every trader at once, so these are descriptive compositions and not treatment effects.

5.3 The pricing rule at each leg

The third alternative is not about currencies at all. It is about the exchanges that price them, because AMMs do not price every pair by the same rule. The constant-product rule sets a price from pool reserves alone and moves it with trade size relative to depth, so it charges a trade for the price impact it causes whether or not the two assets are close substitutes. Curve's invariant instead interpolates toward a flat curve in the neighbourhood of parity, which is exactly where two dollar-pegged assets trade, and Appendix B.3 states the interpolation the venue applies. On this account the rotation is an adoption of technology: a leg through a stablecoin became cheap once venues priced pegged assets on a curve built for them, and the vehicle role passed to stablecoins because of where their legs could be priced, so no monetary property of the asset need enter the account at all. The timing is favourable to it, since stable-specialised pools grew over the same years as the rotation.

The account can be tested by restricting attention to routes on which the pricing rule never changed. A route component enters the constant-product scope only when every one of its legs executes at Uniswap v1 through v4 or at SushiSwap v2 or v3, each of which prices locally on the product of reserves, concentrated liquidity inside an active tick included. No leg inside that scope can have been priced by a curve specialised for pegged assets, so a technology account predicts that stablecoins gain nothing there. The quantity compared is the excess-use ratio of Section 3.3, an asset's share of intermediary activity divided by its share of route-endpoint demand, formed separately inside each scope, so that the denominator for a venue family is the endpoint demand observed in that same family.

The restriction does not weaken the rotation; it sharpens it. Where every leg is priced by the reserve product, stablecoin excess use rises from 0.78 in 2024 to 1.33 in 2026 by value, a movement of +0.55 against +0.40 over all venues, and the ratio crosses one inside the restricted scope in the same comparison. Native assets fall to 0.62 there over the same years. By count, where stablecoins were already over-represented as intermediaries at the start of the sample, the restricted and unrestricted paths are almost the same: 1.28 to 1.42 inside the scope against 1.27 to 1.41 over all venues. The scope is not a small corner of the sample either, since routes priced end to end by the constant-product rule carry 84.8% of candidate-currency intermediary episodes in 2026. Table 8 reports every year and every venue family.

The venue that embodies the alternative supplies no intermediation at all. In each year of the sample, route components all of whose legs execute on Curve contain no intermediary episode whatever. The stable-specialised curve is used to exchange two pegged assets with one another, not as a place through which a route passes on its way somewhere else, so the pricing rule that makes stablecoin legs cheap operates on the direct trades that the vehicle measure excludes by construction. Curve legs do appear inside routes that also touch another exchange, and those routes are counted in the all-venue scope; what the sample contains no instance of is a route intermediated entirely within the specialised invariant. Balancer, whose pools mix weighted and stable families and which the panel therefore leaves technologically unlabelled, carries 39,536 intermediary episodes in its busiest year against 8,767,213 in the busiest year of the constant-product scope, and its ratios swing accordingly; it is reported as a composition diagnostic and no weight rests on it.

What the comparison licenses is again bounded. Holding the pricing rule fixed by restriction is not an estimate of its effect: venues are chosen alongside routes, and a trader who would have used a stable-specialised pool may appear in the constant-product scope only because that pool was unavailable for the pair at hand. The ratio's denominator removes the most obvious composition problem, since each scope is judged against the endpoint demand observed inside it, but it cannot remove selection into scopes. The conclusion is the narrower one that the rotation is not an artifact of stable-specialised pricing, because it is present, large, and in the same direction where that pricing was never applied.

5.4 Liquidity in the vehicle's own pools

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). On a decentralised exchange, the relevant markets are the pools connecting an intermediary to other assets. Trading can attract provider capital, deeper pools can lower price impact, and lower price impact can draw further trading. Pool size remains an equilibrium outcome because providers balance fee income against adverse-selection risk and the cost of adjusting their positions (Lehar and Parlour, 2025). Deposited capital measures funds committed to a pool; a size-specific quote measures how much liquidity is executable within a stated price impact. The distinction matters. Concentrated-liquidity providers can alter executable depth by changing where capital

Table 8: Vehicle excess use by venue pricing family

	All venues		Constant product		Balancer		Curve	
Year	Count	Value	Count	Value	Count	Value	Count	Value
Panel A: Stablecoins								
2020	1.45	0.42	1.50	0.42	no routes		no intermediation	
2021	1.18	0.89	1.20	0.92	2.09	1.43	no intermediation	
2022	1.46	1.08	1.50	1.12	1.91	1.00	no intermediation	
2023	1.89	0.94	1.95	1.00	1.46	0.37	no intermediation	
2024	1.27	0.84	1.28	0.78	1.01	0.27	no intermediation	
2025	1.23	1.21	1.23	1.20	0.70	0.09	no intermediation	
2026	1.41	1.24	1.42	1.33	1.75	1.68	no intermediation	
Panel B: Native asset								
2020	0.91	1.56	0.90	1.37	no routes		no intermediation	
2021	0.94	1.13	0.94	1.09	0.24	0.66	no intermediation	
2022	0.85	0.96	0.85	0.93	0.35	0.90	no intermediation	
2023	0.87	1.16	0.88	1.08	0.50	0.53	no intermediation	
2024	0.95	1.20	0.95	1.29	0.22	0.20	no intermediation	
2025	0.89	0.67	0.92	0.85	0.19	0.19	no intermediation	
2026	0.77	0.58	0.80	0.62	0.51	1.30	no intermediation	

Note: Excess use is an asset type's share of intermediary activity divided by its share of route-endpoint demand, formed over the candidate currencies within each scope. A route component enters a scope only when every one of its legs executes at a venue in that family. The constant-product family is Uniswap v1 through v4 and SushiSwap v2 and v3. Balancer is listed separately and left technologically unlabelled because its pools mix weighted and stable families. Value ratios use all routes; the 20% value-agreement support of Table 2 is narrower. Balancer has no route components before 2021, and 2026 covers January through June.

is placed along the price curve, so deposited capital and executable liquidity coincide only under particular pool designs (Adams et al., 2021).

That mechanism operates inside a market. Deeper pools joining stablecoins to other assets reduce the price impact of carrying a given source–destination trade through a stablecoin, so the classical account asks the trader to switch intermediary in a pair that was already being traded. Section 3.2 measures exactly that margin, and it is the one that stays still: within pairs traded in both years, the stablecoin share changes by +0.2 pp by count, and the interval around it excludes within-pair increases beyond +1.7 pp. Depth in stablecoin pools cannot therefore carry the rotation through continuing markets. What it can still do is act earlier, by making a stablecoin the asset to which a newly traded or fast-growing pair is connected when its markets are first formed. Separating the two roles calls for the capital committed to each candidate's pools on the same daily calendar as the routes, read in each direction on its own, because a contemporaneous association between trading and depth is equally consistent with either.

The local bridge-depth screen supports that earlier-margin interpretation. For each source–destination–date–scope opportunity in which at least two of the five headline candidates have both atomic legs supported, we rank candidates by the weaker prior-calendar deposited-capital leg in the source–candidate–destination bridge. The deepest supported bridge carries 84.1% of five-candidate route mass, rising from 75.5% in 2024 to 86.5% in 2026, across 58,483 opportunity cells and 2,923 ordered pairs. With opportunity and candidate effects, a one-log-point larger weaker-leg capital stock is associated with +6.63 pp higher route share, with standard error 1.22 pp; the corresponding stable-candidate total slope is +7.69 pp, with standard error 1.46 pp. Adding same-day leave-one-out candidate route count and prior-30-day candidate reach leaves the local-depth coefficient at +6.74 pp, with standard error 1.15 pp; same-day global route reach contributes +9.04 pp, with standard error 2.80 pp. Restricting the same candidate set to first-observed pair dates, and retaining zero-depth alternatives as part of the birth choice set, gives the entry margin directly: across 835 candidate rows in 167 entry opportunities, a one-log-point larger local bridge predicts +5.39 pp higher route share and +5.32 pp higher selection probability, with standard errors 0.52 pp and 0.57 pp. Prior-30-day route reach is estimated at +9.35 pp with standard error 5.76 pp, so the birth result is not simply that already popular vehicles get chosen. A two-leg bottleneck screen gives the physical-market version of the mechanism: holding the stronger leg and reach fixed leaves the weaker-leg slope at +6.57 pp, with standard error 1.17 pp, while a one-log-point leg imbalance predicts −5.42 pp route share, with standard error 1.14 pp. The local-depth slope also survives all 5 leave-one-candidate reruns of the same horse race, with the weakest retained slope +5.96 pp and standard error 1.28 pp, so the result is not driven mechanically by any single headline vehicle. Inside the stable-only supported race, 12,048 opportunity cells across 296 ordered pairs compare DAI, USDC, and USDT directly. With the same opportunity effects, local depth, and reach controls, the 2026 USDC interaction is +34.71 pp with standard error 17.20 pp, the USDT interaction is +55.38 pp with standard error 13.17 pp, and local depth remains +5.96 pp with standard error 1.95 pp. The stablecoin result is therefore not only a generic dollar-asset result; by

2026 the within-dollar race is concentrated in the two operational dollar vehicles. This does not turn deposited capital into a treatment. It says route choice is organised around both issuer-specific use and corridor-specific two-leg depth exactly where market formation and corridor growth give the vehicle role room to be made.

A dynamic bridge screen gives the feedback version of the volume–cost mechanism in Krugman (1980). Matching the same source–candidate–destination bridge at exact future horizons, a 10 percentage point higher current route share predicts +0.020 log points more bridge-depth growth after 30 days and +0.097 after 120 days, with standard errors 0.005 and 0.013. In the other direction, a one-log-point larger current bridge predicts +1.26 pp more route-share growth after 30 days and +1.60 pp after 120 days, with standard errors 0.19 pp and 0.27 pp. The feedback margin is stronger for stable candidates: the corresponding 120-day stable route-share to depth-growth coefficient is +0.115 for a 10 percentage point route-share difference, with standard error 0.012, and the stable depth to route-share coefficient is +2.46 pp with standard error 0.29 pp. These rows absorb candidate and origin-date effects and cluster by ordered pair and date. They do not show that liquidity providers fund route demand one for one, but they move the evidence from a static sorting result toward a local feedback margin: route use and corridor depth forecast one another inside the same bridge.

The allocation of capital itself cuts against a flow-chasing interpretation. In the daily five-candidate V2 panel, WETH is the largest deposited-capital candidate on 100.0% of 2,239 days, while stablecoins lead excess use on 81.3% of days. From 2024 to 2026, stablecoins' share of candidate deposited capital falls from 17.6% to 11.1%, while their share of candidate intermediary route counts rises from 18.1% to 46.6%, leaving their 2026 route share 4.2× their capital share. On the matched January-through-June daily calendar, the stablecoin route-minus-capital gap expands by +33.5 pp with standard error 1.3 pp, while the WETH gap moves by −36.6 pp with standard error 1.2 pp. Holding the date fixed, and controlling for endpoint share, venue count, and pool count, stablecoin candidates still carry +13.3 pp more route share than deposited-capital share, with standard error 0.6 pp.

The signed gap predicts later capital reallocation, but gradually: within candidate and origin-date effects, a 10 percentage point route-minus-capital gap predicts +0.61 pp higher candidate capital share after 30 days, with standard error 0.09 pp, and +1.65 pp after 120 days, with standard error 0.11 pp. For stable candidates, the corresponding 30-day signed-gap coefficient is +0.78 pp, with standard error 0.16 pp. The stable response is not homogeneous: at 120 days, the same 10 percentage point gap predicts +5.17 pp more DAI capital share, +0.75 pp more USDC capital share, and only +0.19 pp for USDT, with standard errors 0.77 pp, 0.12 pp, and 0.13 pp. In log deposited capital the USDC and USDT responses are +14.44% and +6.94%. Rank transitions make the same point in a unit closer to portfolio choice: for stable candidates, a 10 percentage point route-minus-capital gap predicts +0.23 positions of capital-rank catch-up after 120 days, with standard error 0.04, while route rank moves −0.27, with standard error 0.02. The response is asymmetric. In a piecewise version of the same design, a 10 percentage point stable capital overhang, meaning capital share exceeds route share by 10 points, predicts a −3.91 pp capital-share change after 30 days and −7.11 pp after 120 days. Aggregating the three stablecoins into a dollar-vehicle basket shows where much of the adjustment comes from. With activity controls and a 30-day calendar HAC, a 10 percentage point stable-basket route-capital shortfall predicts +2.30 pp higher stable-basket capital share after 120 days and −2.02 pp lower WETH capital share, with standard errors 0.45 pp and 0.56 pp; the WBTC response is only −0.28 pp. The extensive margin also fails the simple expansion story. For stable candidates, a 10 percentage point route-minus-capital gap predicts +0.56% future log venue-count change after 30 days and +0.55% after 120 days, but −2.73% future log pool-count change after 120 days. The stable capital that moves is also more concentrated: by 2026 its capital-weighted top-pool share is 52.2%, versus 11.4% for non-stable candidates, and the same 10 percentage point stable gap predicts +2.68 pp higher future pool-capital HHI after 120 days, with standard error 1.02 pp.

The USDC/SVB stress window gives the same separation in event time. Stable route demand rose sharply, with stable basket route share increasing by +11.4 pp inside the five-candidate set, while stable deposited-capital share moved −2.0 pp during the same window and −3.3 pp afterward. USDC's own capital share was −3.2 pp lower after the window, while WETH's was +3.1 pp higher. The episode combines peg stress, arbitrage, and demand, so it remains descriptive. It still falsifies a mechanical capital-chases-current-routes story: the route shock is visible immediately, while the V2 capital stock shifts away from stablecoins over the same event horizon.

A same-pool screen gives the rival its best shot. Across pool-candidate rows, a 10 percentage point route-minus-capital gap predicts +2.36% future log deposited-capital growth after 120 days, with standard error 0.46%; for stable candidates the corresponding total is only +0.72%, with standard error 0.87%. Splitting future exact-state V2 capital by whether the pool was already active at the origin date gives a sharper boundary: stable shortfalls predict +12.39% incumbent-pool capital after 120 days, with standard error 5.42%, but entrant-pool capital and total capital are +1.06% and +4.39%, with standard errors 1.92% and 2.69%, and are not statistically clean in this specification. Rent incidence gives the same answer in the V3 fee archive: across 208,360 same-pool horizon rows in 3,034 pools, a 10 percentage point stable route-minus-capital gap predicts +0.10% future log fees and +1.53% future log volume after 120 days, with standard errors 0.78% and 1.11%. Normalizing by TVL leaves no hidden rent-yield story: the corresponding estimates for future log fee yield and future log volume turnover are −0.27% and 0.00%, with standard errors 0.46% and 0.21%. Provider actions do move on a different margin. In 11,195 candidate-days built from Uniswap v3 mint and burn logs, a 10 percentage point stable route-minus-capital gap predicts +67.89% more future mint events and +64.20% more burn events over 30 days, while the net mint-minus-burn event balance rises only +1.82 pp, with standard errors 3.70%, 3.53%, and 0.19 pp. The provider-day version points the same way: total future mint-or-burn origin counts rise +59.11%, with standard error 3.26%. Net of the candidate's own origin-day V3 action and provider-day counts, the provider-day estimate remains +4.81% at 30 days and +6.02% at 120 days, with standard errors 0.71% and 0.95%. Splitting the same design by issuer shows that the churn is not uniform: total V3 LP actions rise by +86.61% for DAI, +41.96% for USDC, and +99.89% for USDT. For USDT, the action response comes from both sides of the book, with future mint events up +98.64% and burn events up +100.25%, and the provider-day response is +91.94%, yet the net event balance is negative at −0.66 pp; DAI and USDC have positive net balances, +5.27 pp and +2.38 pp. The V4 modify-liquidity archive sharpens the protocol boundary. Among 1,225,274 five-candidate V4 candidate-action assignments, stablecoins account for 56.6%, compared with 31.4% in the V3 mint/burn assignment panel; USDT alone is 22.6% in V4, compared with 11.3% in V3. V4 actions are range-management intensive: 69.3% of assignments are narrow or medium tick ranges, while only 4.8% are full-range. Those are event and provider-day counts, not dollar-valued provider flows or unique LP identities, and the V4 comparison maps native ETH to the WETH candidate family. Within candidate and origin-date effects, a one log-point higher capital stock is associated with +0.90 pp higher intermediary episode share and +0.50 higher excess-use ratio. These associations say capital stocks, venue reach, LP action counts, and vehicle roles are organised together, but that liquidity provision adjusts more like inventory rebalancing, range-management churn, and venue-footprint selection than immediate funding of every stable route-demand shortfall in the same pool, consistent with constrained liquidity provision in currency markets (Huang et al., 2025); they do not show provider flows causing the rotation.

The forward V4 exercise uses the singleton's own swap and modify-liquidity logs. Within candidate and origin-date effects, a 10 percentage point stable route-minus-capital gap predicts +5.40 pp higher multi-leg V4 transaction share, +4.50 pp higher same-candidate internal-leg share, and +2.32 pp higher gross-to-net singleton reduction, with standard errors 0.57 pp, 0.42 pp, and 0.40 pp. The corresponding LP-flow specification values each candidate token side of V4 modify-liquidity events. On V4-active origin days, net of current candidate-side flow activity, the same gap predicts +30.91% more future admissible LP flow after 120 days, with standard error 4.09%. Both sides of the book move: add flow rises +23.99%, with standard error 3.64%, while remove flow rises +40.07%, with standard error 4.74%. Net of the candidate's current V4 action, origin-count, and sender-count activity, the same gap predicts +30.04% more future V4 modify-liquidity actions and +12.64% more future sender-days. This is the liquidity-provision result connected to the V4 accounting design: the economic route remains visible even when the singleton nets settlement, and LPs respond on flow, action-count, and sender-day margins (Lehar and Parlour, 2025). It is still filtered candidate-side flow, not whole-pool TVL, a true LP inventory, or a causal provider response.

Table 9 consolidates the route-capital-gap regressions. A compact notation for the candidate-day rows is

$$Y_{c,t+h} = \beta G^{S}_{c,t} + X'_{c,t}\Gamma + \alpha_c + \delta_t + \varepsilon_{c,t+h}, \tag{15}$$

where $G^{S}_{c,t}$ is the stable-candidate route-minus-capital gap at the origin date. The table reports the effect of a 10 percentage point larger gap. The estimate is positive for gradual capital adjustment and for incumbent-pool capital, but it is small and insignificant in same-pool capital and V3 fee yield. V3 responds through mint-and-burn churn, and V4 responds through flash-accounting intensity, candidate-side LP flow, actions, and sender-days, which is the provider-side margin closest to constrained-liquidity adjustment in Huang et al. (2025).

Table 10 reports the corresponding route-to-provider mechanism in the opposite direction, using the singleton's own accounting intensity at the candidate-day level:

$$Y_{c,t+h} = \beta M_{c,t} + X'_{c,t}\Gamma + \alpha_c + \delta_t + \varepsilon_{c,t+h}. \tag{16}$$

Here $M_{c,t}$ is one of the three flash-accounting proxies, $X_{c,t}$ contains origin-day swap activity, current candidate-side LP flow, current candidate-linked TVL, current LP actions, and the current narrow-or-medium range share, and the fixed effects compare candidates on the same day. The table uses the 120-day horizon because that is the interval over which the earlier V2 and V3 provider responses are most visible. Internal same-candidate settlement and multi-leg routing forecast more future LP flow, more candidate-linked TVL, and more LP actions. The position result is not tightening: those same proxies predict lower narrow-or-medium shares and higher wide-or-very-wide shares, so activity moves toward broader repositioning rather than more concentrated placement.

Table 11 joins the two margins. The row variable is the interaction between the stable route-minus-capital gap and the same-day V4 accounting proxy, with the same candidate and date effects and the same origin-day activity controls as Table 10. The estimates separate two parts of the singleton design. When a stable candidate has both a larger shortfall and more internal or multi-leg singleton routing, future candidate-linked TVL and LP actions rise, and future LP placements move away from narrow-or-medium ranges toward wide-or-very-wide ranges. Gross-to-net reduction moves the other way: it predicts lower future TVL and actions, more narrow-or-medium placement, and less wide-or-very-wide placement. The result says that the economically visible V4 route margin is connected to provider reallocation, but not because every form of net settlement compression has the same liquidity meaning. That distinction fits a view of liquidity providers as active inventory and range managers facing fee, adverse-selection, and adjustment-cost tradeoffs (Lehar and Parlour, 2025; Huang et al., 2025).

Table 12 makes the protocol comparison directly on the same candidate and day. It restricts the sample to origins with observed V4 singleton swap activity, stacks the V3 mint/burn outcome and the V4 modify-liquidity outcome, absorbs candidate-date and protocol fixed effects, and controls for the same protocol's current LP actions and active origins. A 10 percentage point stable route-minus-capital shortfall is associated with a V4-minus-V3 future LP-action differential of 0.320 log points after 120 days and a corresponding active-origin differential of 0.115 log points. The comparison does not estimate the causal effect of adopting V4. Its narrower role is to rule out a reading in which V4 merely has a higher level of counted activity: conditional on the same candidate-date origin, the stable shortfall is more strongly linked to future provider action in V4 than in V3.

Table 9: Liquidity-provision responses to stable route-capital gaps

Margin	Outcome	Horizon	Effect	Unit	Obs.	Clusters
V2 stock	Capital share change	120d	$+1.4^{***}$ (0.2)	pp	10,595	2,119
V2 stock	Log deposited capital	120d	$+0.205^{***}$ (0.031)	log pts	10,595	2,119
Portfolio rank	Capital-rank catch-up	120d	$+0.225^{***}$ (0.036)	ranks	10,595	2,119
Same pool	Same-pool capital	120d	+0.007 (0.009)	log pts	174,712	2,113
Pool footprint	Incumbent-pool capital	120d	$+0.124^{**}$ (0.054)	log pts	10,417	2,117
Rent incidence	V3 fee yield	120d	−0.003 (0.005)	log pts	208,360	189
V3 churn	Mint events	30d	$+0.679^{***}$ (0.037)	log pts	11,195	2,239
V3 churn	Burn events	30d	$+0.642^{***}$ (0.035)	log pts	11,195	2,239
V3 churn	Net mint-burn balance	30d	$+0.018^{***}$ (0.002)	events	11,195	2,239
V3 activity-controlled	Provider-day activity	120d	$+0.060^{***}$ (0.010)	log pts	11,195	2,239
V4 accounting	Multi-leg transaction share	same day	$+5.4^{***}$ (0.6)	pp	2,599	522
V4 accounting	Internal same-candidate share	same day	$+4.5^{***}$ (0.4)	pp	2,599	522
V4 accounting	Gross-to-net reduction share	same day	$+2.3^{***}$ (0.4)	pp	2,599	522
V4 LP flow	Gross candidate-side flow	120d	$+0.309^{***}$ (0.041)	log pts	2,592	521
V4 LP flow	Add flow	120d	$+0.240^{***}$ (0.036)	log pts	2,592	521
V4 LP flow	Remove flow	120d	$+0.401^{***}$ (0.047)	log pts	2,592	521
V4 activity-controlled	LP actions	120d	$+0.300^{***}$ (0.024)	log pts	2,599	522
V4 activity-controlled	Sender-days	120d	$+0.126^{***}$ (0.016)	log pts	2,599	522

Note: Each cell reports the effect of a 10 percentage point higher stable-candidate route-minus-capital gap, with clustered standard errors in parentheses. Units are percentage points, log points, rank positions, or event balance as indicated. Candidate-day rows absorb candidate and origin-date effects unless the row label implies a pool-level design: same-pool capital absorbs pool-candidate and origin-date effects, V3 fee yield absorbs pool and origin-date effects, and V4 rows use V4-active origin candidate-days. V3 activity-controlled and V4 activity-controlled rows include current origin-day action, origin-count, and sender-count controls; V4 flow rows include current candidate-side LP-flow controls. These are predictive mechanism regressions, not causal provider-supply estimates. Stars denote

Table 10: V4 flash accounting and subsequent liquidity provision

Flash-accounting proxy	Future LP flow (log pts)	Candidate-linked TVL (log pts)	LP actions (log pts)	Narrow/medium ranges (pp)	Wide/very-wide ranges (pp)
Internal same-candidate share	$+0.035^{*}$ (0.020)	$+0.265^{***}$ (0.031)	$+0.071^{***}$ (0.013)	-2.460^{***} (0.384)	$+2.889^{***}$ (0.354)
Multi-leg transaction share	$+0.076^{**}$ (0.032)	$+0.123^{***}$ (0.018)	$+0.058^{***}$ (0.017)	-1.867^{***} (0.203)	$+1.808^{***}$ (0.224)
Gross-to-net reduction share	$+0.139^{***}$ (0.041)	+0.026 (0.031)	$+0.044^{**}$ (0.019)	+0.362 (0.373)	+0.021 (0.328)
Observations / date clusters					2,015 / 403
Candidate and date effects					Yes
Origin-day activity controls					Yes

Note: Each cell reports the effect of a 10 percentage point increase in the row proxy on the column outcome, with standard errors in parentheses. The unit is a candidate-day in Uniswap v4. All columns use the 120-day horizon, absorb candidate and origin-date fixed effects, include origin-day swap activity, current candidate-side LP flow, current candidate-linked TVL, current LP actions, and current range-composition controls, and cluster by origin date. LP flow is the admissible dollar value of candidate-token sides in modify-liquidity events. Candidate-linked TVL is full-pool provider TVL linked to a candidate token side after address-resolution and physical TVL/volume bounds. Range shares partition future modify-liquidity events into narrow-or-medium, wide-or-very-wide, and full-range placements. Stars denote 1%, 5%, and 10% significance.

Table 11: Stable shortfalls, V4 flash accounting, and LP repositioning

Interaction term	Candidate-linked TVL (log pts)	LP actions (log pts)	Narrow/medium ranges (pp)	Wide/very-wide ranges (pp)
Stable gap × internal same-candidate share	+0.292*** (0.079)	+0.082*** (0.027)	−6.531*** (0.720)	+6.189*** (0.657)
Stable gap × multi-leg transaction share	+0.225*** (0.035)	+0.132*** (0.013)	−4.539*** (0.442)	+4.438*** (0.381)
Stable gap × gross-to-net reduction share	−0.667*** (0.105)	−0.121*** (0.043)	+4.411*** (1.261)	−4.405*** (1.126)
Observations / date clusters				1,209 / 403
Candidate and date effects				Yes
Origin-day activity controls				Yes

Note: Each cell reports the interaction of a 10 percentage point larger stable route-minus-capital gap with a 10 percentage point larger V4 accounting proxy. Log outcomes are in log points; range outcomes are in percentage points. The unit is a stable candidate-day in Uniswap v4. All columns use the 120-day horizon, absorb candidate and origin-date fixed effects, include origin-day swap activity, current candidate-side LP flow, current candidate-linked TVL, current LP actions, and current range-composition controls, and cluster by origin date. Candidate-linked TVL is full-pool provider TVL linked to a candidate token side after address-resolution and physical TVL/volume bounds. Range shares partition future modify-liquidity events into narrow-or-medium, wide-or-very-wide, and full-range placements. The design is exploratory mechanism evidence, not causal LP-supply identification. Stars denote 1%, 5%, and 10% significance.

Table 12: Stable shortfalls and the V4-minus-V3 LP-action response

Outcome	7 days	30 days	120 days
LP actions	+0.163***	+0.212***	+0.320***
	(0.016)	(0.015)	(0.013)
Active origins	+0.045***	+0.062***	+0.115***
	(0.011)	(0.011)	(0.010)
Stacked observations / dates			5,198 / 522
Candidate-date and protocol effects			Yes
Current protocol LP controls			Yes

Note: Each cell reports the V4-minus-V3 difference in the association between a 10 percentage point larger stable route-minus-capital gap and the future LP-action outcome, with standard errors in parentheses. Outcomes are log points. The stacked unit is a protocol–candidate–origin-day row, restricted to candidate-origin days with observed V4 singleton swap activity. All columns absorb candidate-date and protocol fixed effects, include same-protocol current LP-action and active-origin controls, and cluster by origin date. V3 outcomes are mint/burn event counts and active origins. V4 outcomes are modify-liquidity event counts and active origins. The design is an exploratory protocol contrast, not a causal protocol-adoption estimate. Stars denote 1%, 5%, and 10% significance.

5.5 Protocol architecture

Protocol design changes the set of routes that can be formed. Uniswap v1 created pools between an ERC-20 token and ETH, so token-to-token trades crossed ETH; Uniswap v2 allowed arbitrary ERC-20–ERC-20 pools and made direct token-to-token trading possible.[7] Forced intermediation was material while the constraint lasted: token-to-token trades routed through ETH account for 8.0% of Uniswap v1 swap transactions and 8.1% of its ETH volume over the venue's life, and roughly one trade in ten in 2019 and 2020, when the venue was most active. A transaction from the crash weekend of March 2020 shows the mandate operating: with no direct pool between MKR and DAI, a trader selling 250 MKR received 439.13 ETH at one exchange contract and spent the identical ETH amount at a second for 49,892.40 DAI.[8] Native-asset pairing remained pervasive after this mandate ended. The share of single-leg Uniswap v2 trades executed in WETH pools was 95.1% in 2020 and 95.5% in 2026, remaining between 93.4% and 97.9% in every intervening year. Among the 477,633 pairs that ever traded on Uniswap v2, 97.1% include WETH, and WETH's share of newly traded pairs rose from 84.1% in 2020 to 97.9% in 2026. This persistence is consistent with deep WETH pools, launch conventions, and inherited search paths. Separating those forces from the original protocol constraint requires comparison among pairs that had the same alternatives available.

Uniswap v4 separates route choice from physical settlement more sharply. All pools reside in one singleton, and flash accounting records balance changes internally before settling the final net

[7] See Adams, Zinsmeister, and Robinson, *Uniswap v2 Core*, pp. 1–2.

[8] Transaction `0x4dca160d…96a0ca16`, block 9,674,728, March 15, 2020. The trace is selected by a deterministic rule, the largest ETH amount routed among the 129,810 mandate-era token-to-token transactions whose two ETH legs match exactly, and the token identities are verified against the public transfer record.

amounts at the end of a call.[9] An economic route can therefore cross an intermediary pool without generating an external token transfer at every step. Decoded pool actions preserve the trading path despite this netting, while the architecture changes how liquidity, fees, and settlement are organised around that path.

The route evidence leaves an explanation six features to account for. The stablecoin gain occurs mainly across pairs rather than through repeated substitution within matched pairs; it appears on short and long paths alike and on routes that stay within one exchange as well as routes that do not; the incidence of intermediation did not step up when the software that searches for paths was rewritten, though the exchange set it reached did widen; the gain is present where every leg was priced by the same unchanged constant-product rule, and absent from the venue whose invariant is built for pegged assets, which carries no intermediation at all; local two-leg depth predicts which candidate carries route mass inside an opportunity; and the measured WETH-pairing persistence shows that route opportunities can outlast the protocol rule that first created them. Together these narrow the candidates. An explanation cannot rest on the composition of the routes that were built, on the software that selected them, on the technology that priced them, or on repeated intermediary substitution in established markets. It has to say why the markets forming and growing over this period attach themselves to a stablecoin.

[9] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

6 Conclusion

Vehicle dominance on decentralised exchanges has shifted markedly from native assets toward stablecoins through a reorganisation of the trading network, with little change of intermediary inside comparable trades. Stable-for-native substitution within matched source–destination pairs is close to zero; the aggregate moves with activity reallocating toward pairs that already route through stablecoins and with pairs observed in only one of the two comparison years. Native assets regain share in 2023 and 2024 before stablecoins recover the value lead, and USDT's intermediary use rises relative to its appearance at the observed endpoints of pool routes.

Vehicle-currency theories usually emphasise a reinforcing choice for a given ultimate pair of assets: trading deepens an intermediary's markets, lower costs attract more trading, and the vehicle role persists. The decomposition adds the network distribution of trading to this account. A vehicle gains aggregate share when it becomes more common within an ultimate pair, when activity shifts toward ultimate pairs that already use it, or when newly active ultimate pairs route through it. In our sample, the latter two margins dominate. This distinction changes how aggregate vehicle shares should be interpreted: a rising share can reflect a reorganisation of trading toward parts of the network in which the challenger already connects markets, even when otherwise comparable trades do not widely replace the incumbent intermediary. Models and empirical tests that hold the ultimate pair fixed describe the first margin; aggregate turnover combines both.

The mechanism evidence locates where such a role is made. Stable vehicles appear in new and repeatedly active corridors. When several headline vehicles are locally feasible, the candidate with the deepest prior two-leg bridge carries most route mass. Liquidity provision moves on a slower margin: deposited capital remains native-heavy, stable-route shortfalls predict later rebalancing and more concentrated pool capital, and the USDC/SVB window shows stable-route demand rising while capital shifts toward WETH. Route opportunities, local depth, identity persistence, and inventory reallocation therefore move together without implying that LP capital immediately funds each route-demand shock.

The accounting is not specific to decentralised exchanges. A currency's share of settled trade can decline while every corridor that uses it today continues to do so, provided trade grows fastest along corridors that never adopted it. The renminbi arrangements between China and Brazil described in Section 1 illustrate one such corridor. Aggregate currency shares are always measured over a set of corridors that is itself changing, and the two margins quantified here are the margins through which such a decline would run. Records that report only the legs of an exchange cannot separate them from a change of intermediary within a corridor. A transaction log that preserves the whole path can.

The economic importance of a vehicle comes from the trading it connects. Vehicle dominance is therefore jointly determined by the intermediary selected for each pair and by the distribution of routed activity across the network.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol's factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, atomic trades (pool-swap legs) are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one ultimate trade between its endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. The output records the sequence of atomic pairs and pools traversed, together with the ultimate pair. It does not infer that sequence from the transaction sender or router label.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. A failed valuation removes the observation from the value calculation while preserving the observed route. Appendix F gives the separate exclusion for self-returning paths.

Historical token precision cannot be verified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool's state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve's amplification coefficient and Balancer's weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the transaction.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p's input- and output-token reserves immediately before transaction e, Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \qquad \varphi_p = \tfrac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.[10]

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated routes at the same state and dollar input notional,

$$10^4 \Delta C^D_{i,o,k,q,e} = 10^4 \frac{O^D_{i,o,q,e} - O^I_{i,o,k,q,e}}{O^D_{i,o,q,e}}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\widetilde{R}_{i,p,e^-}$ and $\widetilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\widetilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \qquad \widetilde{R}_{o,p,e^-} = L_{p,e^-}\sqrt{\rho_{p,e^-}}.$$

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \widetilde{R}_{o,p,e^-}}{\widetilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

[10] The SushiSwap v2 pair contract implements the same adjustment; see `UniswapV2Pair.sol`.

Table 13: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine pairs, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

until the trade reaches the next initialized tick. Range boundaries lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \qquad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract's integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrtPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 13 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.[11] The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing group reproduces realised output within 0.001% at the worst observed swap.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.[12] For n tokens with normalized pre-transaction balances

[11] See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

[12] See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

R_{j,p,e^-}, amplification A_p, and invariant D_p,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \to 0$ recovers the constant product and $A_p \to \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i. Defining $D_\Pi = D_p \prod_j D_p/(nR_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + nD_\Pi)D_p}{(A_p n - 1)D_p + (n+1)D_\Pi}, \qquad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is off by a factor of 10^{12}. Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1)\varphi_p,$$

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it, so it is identified from data instead of fetched. Every other quantity in the invariant is observed and every Curve swap reports both amounts, which leaves A_p as the only unknown,

$$\widehat{A}_p = \arg\min_{A_p} Q_{0.90}\left\{\frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o}\right\},$$

fitted on the first half of a pool-day's trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this component exercise.

Table 14 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

Table 14: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\widehat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\widehat{A}_p$ is reported in the module's convention, which carries the on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,[13]

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \qquad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pair with the same reserves therefore return the same quote. Balancer's weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.[14] The counterfactual design applies the common 5% own-price-impact cutoff in Equation (14) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool's owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which the validation displayed before anything diagnosed it: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p \Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both as unknowns to be recovered from the trades a pool actually executed, the move Curve's amplification coefficient needed in Section B.3, and we keep a reported value only where it survives that test. Reconstructing a pool's state from its own event history follows Lehar and Parlour (2025); what the weighted family adds is that two arguments of the pricing

[13]See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

[14]See the input-ratio bound in Balancer's `WeightedMath.sol`.

function are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day's trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 15 is measured on the evaluation set. For a candidate parameter pair ϑ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90}\left\{ \frac{|Q_p(i \to o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p \right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. The upper tail does the gating for the reason it does at Curve, and the coverage condition stops a parameter that prices one corner of a pool's activity from standing for the pool.

A pair's weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\widehat{\theta}_{ab,p} = \arg\min_{\theta} \mathcal{E}_p\big(\theta \text{ on } \mathcal{O}_p^{a\to b},\ \theta^{-1} \text{ on } \mathcal{O}_p^{b\to a}\big),$$

searched over a logarithmic grid spanning Balancer's own weight bounds and refined by ternary search in the log exponent, and attempted only for pairs carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as a clean fit. The fee $\widehat{f}_p$ is recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its trades on reported parameters is never handed a fitted one,

$$\widehat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \le \bar{e}, \\ (\theta_p, \widehat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \widehat{f}_p) \le \bar{e}, \\ (\widehat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\widehat{\theta}_{\cdot,p}, f_p) \le \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \qquad \bar{e} = 0.1\%. \tag{17}$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the pairs that happened to fit and then quoted on the rest.

How often the machinery binds is itself a diagnostic. Of the 256 pool-days accepted across the twelve validation days, 209 clear the gate on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-pair weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

The gate $\bar{e}$ is set at 0.1% and not at the 1% Curve uses, because the achieved errors separate

Table 15: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day's event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day's opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

cleanly there. The threshold separates weighted pools, whose held-out errors range from 10^{-9} to 0.004%, from stable, composable-stable, and elliptic-curve pools, whose held-out errors range from 0.05% to 1.08% and have a 99th percentile of 12%. At a 1% gate the second group clears on a fitted weight ratio and arrives quoted instead of excluded, which is the Curve lesson in a second form: there the over-permissive object was a parameter range and here it is an error threshold.

Across the twelve validation days, backward reconstruction returns a median error of 0.0000% and identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. The exercise therefore validates the model for identified weighted pools, not for Balancer as a whole.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. The exercises below validate that procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p's stored period-end reserve vector and $\boldsymbol{\Delta}_{p,e}$ the vector of net signed token amounts in event e. The state before event e is the stored value with every later event removed,

$$\mathbf{R}_{p,e}^{\text{pre}} = \mathbf{R}_p^{\text{end}} - \sum_{e' \geq e} \boldsymbol{\Delta}_{p,e'} = \mathbf{R}_{p,e+1}^{\text{pre}} - \boldsymbol{\Delta}_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period's stored value as the current period's opening state and replays forward,

$$\widetilde{\mathbf{R}}_{p,e}^{\text{pre}} = \mathbf{R}_{p,\text{prev}}^{\text{end}} + \sum_{e'<e} \mathbf{\Delta}_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances, so the ordered replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}_{p,1}^{\text{pre}} - \mathbf{R}_{p,\text{prev}}^{\text{end}} \right) \oslash \mathbf{R}_{p,\text{prev}}^{\text{end}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 15 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Component-validation samples contain executed swaps and therefore pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. We include a hypothetical leg in the cost comparison only when its own price impact is at most 5%, calculated under the pricing rule of the pool family it would traverse.

For constant-product swaps, input relative to reserves provides a transparent size–depth diagnostic,

$$\iota_{i,p,e^-} = \frac{\Delta_i}{R_{i,p,e^-}},$$

the input amount as a fraction of the input-side reserve. Table 16 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a

Table 16: Size relative to depth in the validation population

Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

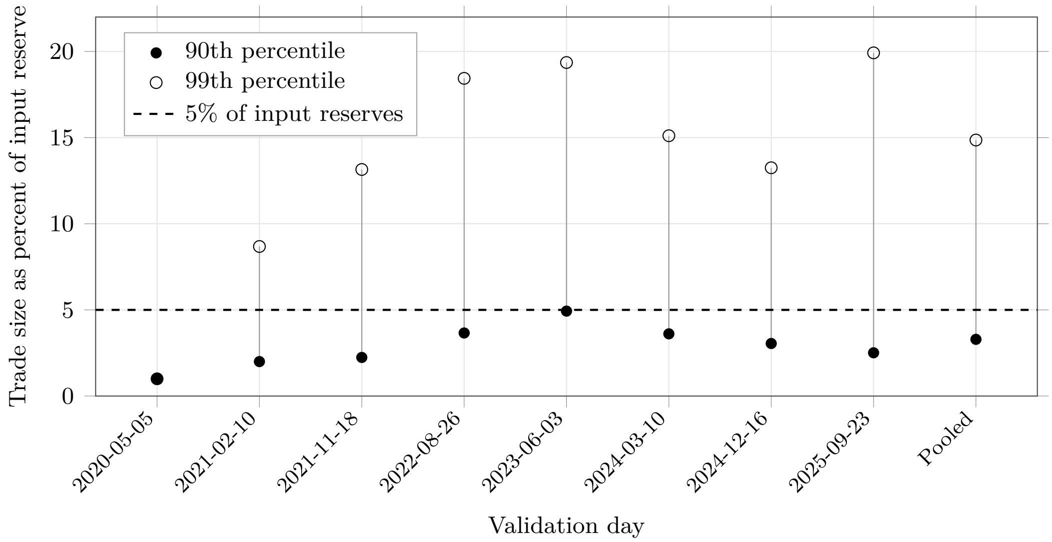
Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quoters were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

Five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86% of the pooled constant-product distribution, and it lies in the same interval on seven of the eight days taken separately. Figure 4 shows the daily distributions. This calculation is a depth diagnostic. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family's pricing function.

Equation (14) applies the same 5% own-price-impact ceiling to every family. The component exercises establish pricing accuracy for realised swaps within their stated samples; they do not measure quote error on unexecuted routes.

Figure 4: Constant-product trade size relative to reserves



Note: The statistic is a leg's input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Omitting an exchange can change the best direct route, the best intermediated route, or both. Table 17 first describes the volume recorded in nine local source series, including Uniswap v1 as a separate architecture sample and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a \$456m day whose swaps sum to \$8, and a constant-product pair with \$255 of reserves reports \$1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 is excluded from the route-cost comparison because its forced-routing architecture is studied separately. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. The table is consequently a source-coverage diagnostic, not an estimate of complete market shares.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits

Table 17: Share of observed decentralised-exchange volume, by source and year

Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61

Note: Percentage shares of volume recorded in nine local source series. The weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Its entries are partial observed-volume contributions, not complete market shares or turnover estimates. The table displays 2020–2026 and pools volume over those years. Uniswap v1 is shown as the separate forced-routing architecture sample and is excluded from the route-cost opportunity set.

621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 percentage points. Table 18 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. The excluded share of native-leg volume exceeds the stable-leg share by at least 33 percentage points in every sample year and by 54 points in 2023. Tricrypto pools pairing WBTC and WETH with a stablecoin account for \$818m, or 44.6%, of the \$1.83bn excluded, including seven of the twelve largest excluded pool-days. These pools use CryptoSwap, so the missing coverage is concentrated where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool's root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only 4.1% of its volume occurs on token pairs unavailable on another included exchange, never more than \$37,609 on one day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023 peak. Table 15 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool's own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (17).

Table 18: Curve volume outside the StableSwap comparison

Panel A: composition of the exclusion					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	$1.038bn	56.6%	15.7%	65.2%
Stable	119	$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	$0.045bn	2.4%	4.4%	22.2%

Panel B: excluded share of each leg's volume, by year			
Year	Native leg	Stable leg	Other volatile
2021	85.49%	46.91%	15.23%
2022	68.82%	30.21%	50.77%
2023	59.11%	5.51%	56.60%
2024	48.24%	0.94%	14.38%
2025	47.47%	13.96%	10.52%
2026	60.92%	9.04%	20.08%

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool claims are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}\{\, k_0 = k_{m(r)} \,\},$$

does not represent a conversion between distinct endpoint assets. We exclude these routes from the dominance measures. The excluded population can contain cyclic arbitrage and other self-returning activity; the endpoint rule does not classify every route as arbitrage or wash trading. Multi-trade arbitrage can be bundled atomically (Daian et al., 2020), but identifying trader intent would require additional evidence.

Table 19 reports the distribution across sampled days; Figure 5 shows the day-level observations.

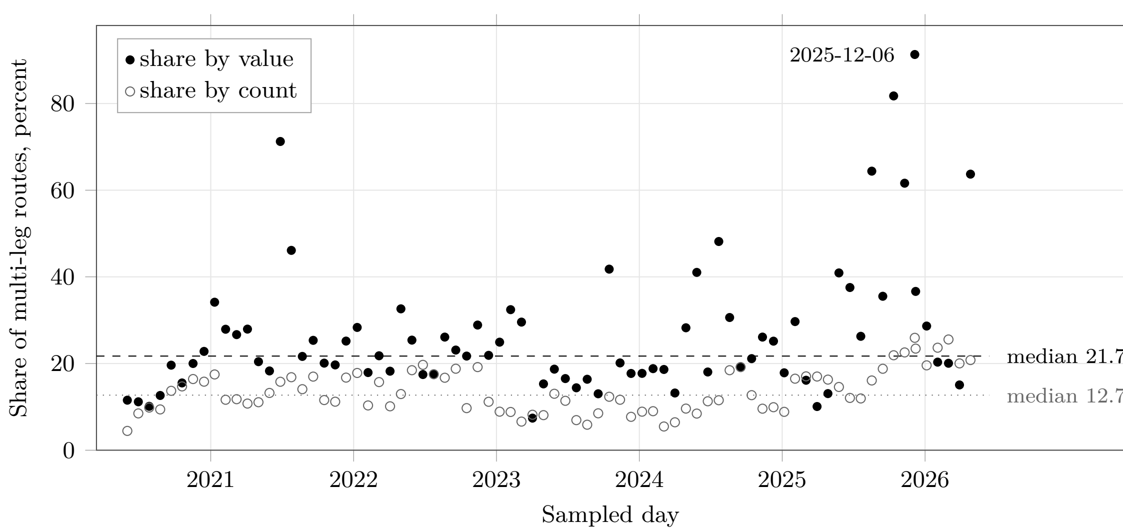
The value-weighted share exceeds the count-weighted share at every quantile, by a factor between 1.7 and 3.5, consistent with a population in which large self-returning routes carry disproportionate value. Round trips appear on every sampled day; their lowest count share, 4.45%, occurs on the earliest date. The paper excludes them before forming both count- and value-weighted dominance measures. Figure 5 plots their count and value shares among multi-leg routes.

Table 19: Round-trip share of multi-leg routing, across 79 sampled days

Quantile across days	Share by count	Share by value
Minimum	4.45%	7.40%
25th percentile	9.60%	17.73%
Median	12.70%	21.72%
75th percentile	17.04%	29.56%
90th percentile	20.03%	46.11%
Maximum	25.93%	91.31%
2020 median	11.77%	14.08%
2021 median	13.21%	25.35%
2022 median	16.72%	21.89%
2023 median	8.51%	17.71%
2024 median	9.60%	21.13%
2025 median	16.75%	36.08%
2026 median	20.80%	20.32%

Note: Round trips as a share of multi-leg routes. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 13,160,047 routes, of which 2,488,399 are multi-leg and 351,553 are round trips. Dates with fewer than 200 multi-leg routes are omitted.

Figure 5: Round-trip contamination of multi-leg routing, day by day



Note: Each mark is one sampled date. *Share by value* weights round-trip routes by their notional; *share by count* gives each route equal weight. Both shares use multi-leg routes as the denominator. Dashed and dotted lines mark the respective medians, and the highest point is labelled. Dates with fewer than 200 multi-leg routes are omitted. The sample contains 79 dates from 1 June 2020 through 27 April 2026 and 2,488,399 multi-leg routes.

References

Adams, H., N. Zinsmeister, M. Salem, R. Keefer, and D. Robinson (2021). Uniswap v3 core. Technical whitepaper.

Amiti, M., O. Itskhoki, and J. Konings (2022). Dominant currencies: How firms choose currency invoicing and why it matters. *Quarterly Journal of Economics 137*(3), 1435–1493.

Anadu, K., P. Azar, M. Cipriani, T. M. Eisenbach, C. Huang, M. Landoni, G. La Spada, M. Macchiavelli, A. Malfroy-Camine, and J. C. Wang (2025). Runs and flights to safety: Are stablecoins the new money market funds? Technical Report Supervisory Research and Analysis Working Paper SRA 23-02, Federal Reserve Bank of Boston. June 2025 revision; first circulated in 2023.

Caparros, B., A. Chaudhary, and O. Klein (2024). Blockchain scaling and liquidity concentration on decentralized exchanges. Working paper, Warwick Business School and Gillmore Centre for Financial Technology.

Catalini, C., A. de Gortari, and N. Shah (2022). Some simple economics of stablecoins. *Annual Review of Financial Economics 14*, 117–135.

Chen, D. and D. Duffie (2021). Market fragmentation. *American Economic Review 111*(7), 2247–2274.

Daian, P., S. Goldfeder, T. Kell, Y. Li, X. Zhao, I. Bentov, L. Breidenbach, and A. Juels (2020). Flash boys 2.0: Frontrunning in decentralized exchanges, miner extractable value, and consensus instability. In *2020 IEEE Symposium on Security and Privacy*, pp. 910–927.

Dowd, K. and D. Greenaway (1993). Currency competition, network externalities and switching costs: Towards an alternative view of optimum currency areas. *The Economic Journal 103*(420), 1180–1189.

Eren, E. and S. Malamud (2022). Dominant currency debt. *Journal of Financial Economics 144*(2), 571–589.

Flandreau, M. and C. Jobst (2009). The empirics of international currencies: Network externalities, history and persistence. *The Economic Journal 119*(537), 643–664.

Gopinath, G., E. Boz, C. Casas, F. J. Díez, P.-O. Gourinchas, and M. Plagborg-Møller (2020). Dominant currency paradigm. *American Economic Review 110*(3), 677–719.

Gopinath, G. and J. C. Stein (2021). Banking, trade, and the making of a dominant currency. *Quarterly Journal of Economics 136*(2), 783–830.

Gorton, G. B. and J. Y. Zhang (2023). Taming wildcat stablecoins. *University of Chicago Law Review 90*(3), 909–971.

Huang, W., A. Ranaldo, A. Schrimpf, and F. Somogyi (2025). Constrained liquidity provision in currency markets. *Journal of Financial Economics 167*, 104028.

Kiyotaki, N. and R. Wright (1989, August). On money as a medium of exchange. *Journal of Political Economy 97*(4), 927–954.

Krugman, P. (1980). Vehicle currencies and the structure of international exchange. *Journal of Money, Credit and Banking 12*(3), 513–526.

Lehar, A. and C. A. Parlour (2025). Decentralized exchange: The uniswap automated market maker. *Journal of Finance 80*(1), 321–374.

Lyons, R. K. and G. Viswanath-Natraj (2023). What keeps stablecoins stable? *Journal of International Money and Finance 131*, 102777.

Makarov, I. and A. Schoar (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics 135*(2), 293–319.

Milionis, J., C. C. Moallemi, T. Roughgarden, and A. L. Zhang (2022). Automated market making and loss-versus-rebalancing.

Mukhin, D. (2022). An equilibrium model of the international price system. *American Economic Review 112*(2), 650–688.

Schär, F. (2021). Decentralized finance: On blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review 103*(2), 153–174.

Somogyi, F. (2026). Dollar dominance in FX trading. *Management Science*.

Xi, W. and C. C. Moallemi (2026). Quantifying sub-optimality in routing for automated market makers. Accepted at the 5th Workshop on Decentralized Finance.